

The image shows a well-lit industrial networking lab. In the foreground, there is a long wooden workbench on a black frame with many drawers. A black ergonomic chair is positioned in front of it. The background is filled with rows of black equipment racks containing various electronic components and wiring. A projector is mounted on the ceiling, and the room has a drop ceiling with fluorescent lights.

INDUSTRIAL NETWORKING

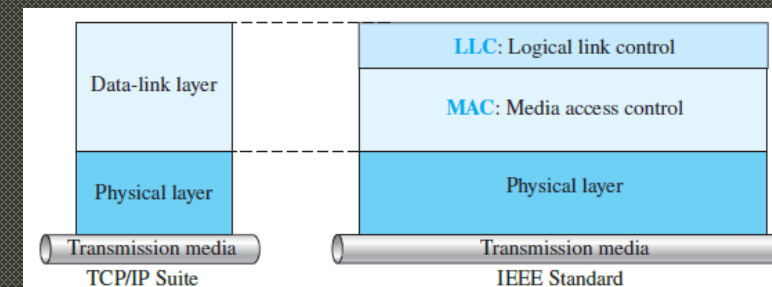
MENG 4010

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6.1 Ethernet

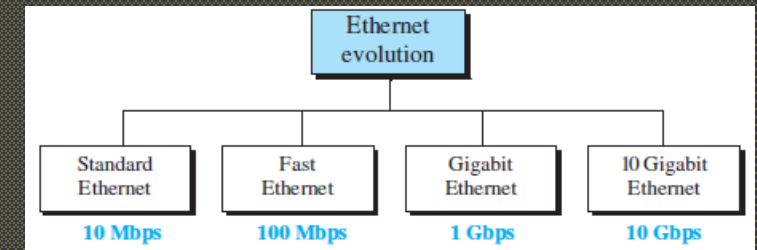
- A LAN connects computers within a small geographic area, like a building, office, or campus.
 - Example: The computers in your school's computer lab connected together.
- Purpose: To share resources such as files, printers, and internet connections.
- Most LANs are now also connected to WANs (Wide Area Networks) or the Internet for external communication.
 - Example: Your school LAN connects to the Internet through a router.
- History of LANs (1980s–1990s)
 - In early days, different types of wired LANs existed (like Token Ring, FDDI, Ethernet). FDDI - Fiber Distributed Data Interface
 - Over time, Ethernet became the standard because it evolved and improved, while others disappeared.
 - Example: Today, when you plug in a network cable, you are using Ethernet - not Token Ring.
- IEEE (Institute of Electrical and Electronics Engineers) sets the standards for LANs.
- They created Project 802 in 1985 to standardize LAN protocols.
 - Purpose: Make sure devices from different manufacturers can work together.
 - Example: A Dell computer and an HP printer can communicate over the same LAN because of IEEE 802 standards.
- Project 802 defines functions for:
 - Physical Layer: cables, signals, connectors.
 - Data-Link Layer: how data is packaged and transmitted.



6.1 Ethernet

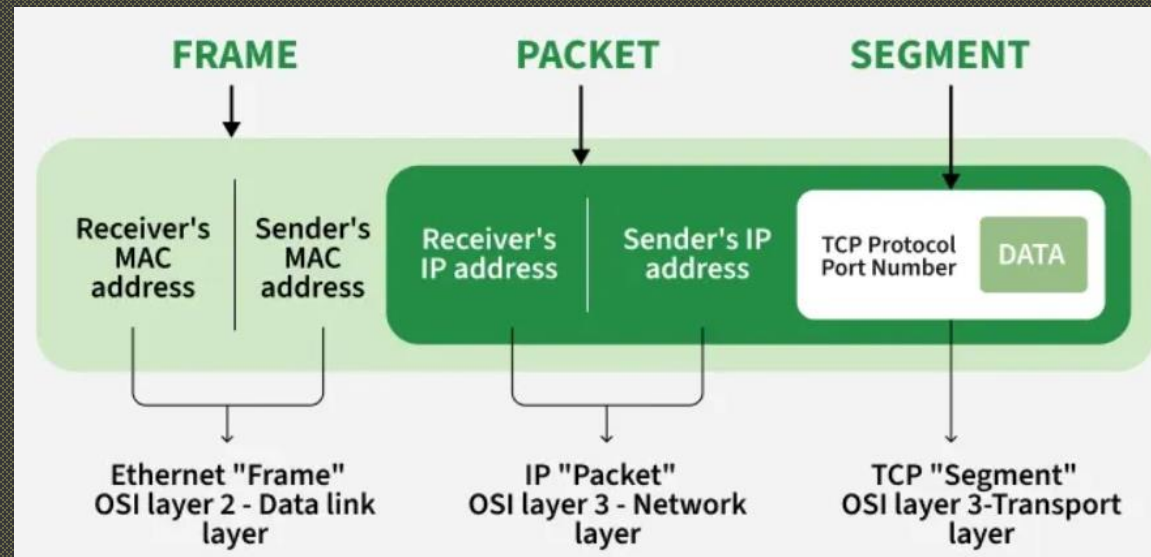
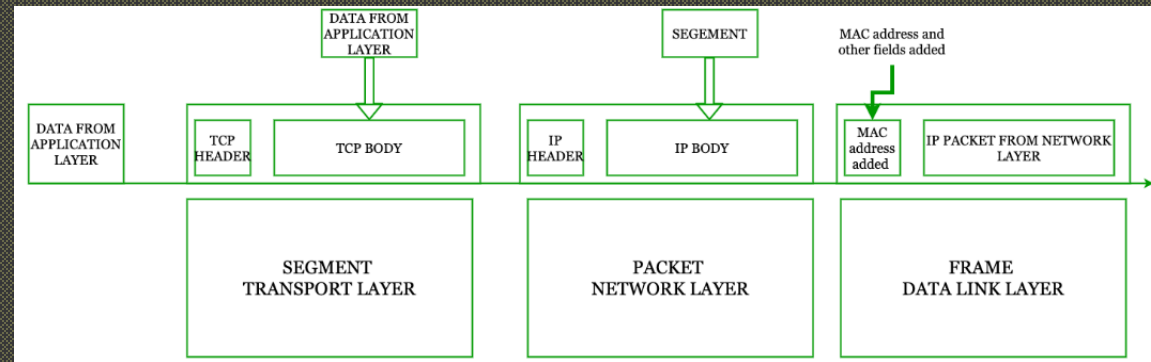
- Data-Link Layer Structure (IEEE 802)
 - The data-link layer is divided into two sublayers:
 - LLC (Logical Link Control):
 - Manages communication between devices.
 - MAC (Media Access Control):
 - Controls how devices share the transmission medium (like a network cable).
 - Example: Decides which computer can send data at a given time to avoid collisions.

- LAN = small area network, WAN = large area (Internet).
- IEEE 802 → sets rules for LAN communication.
- Data-link layer = LLC + MAC.
- Ethernet dominates because it is flexible, fast, and upgradable.

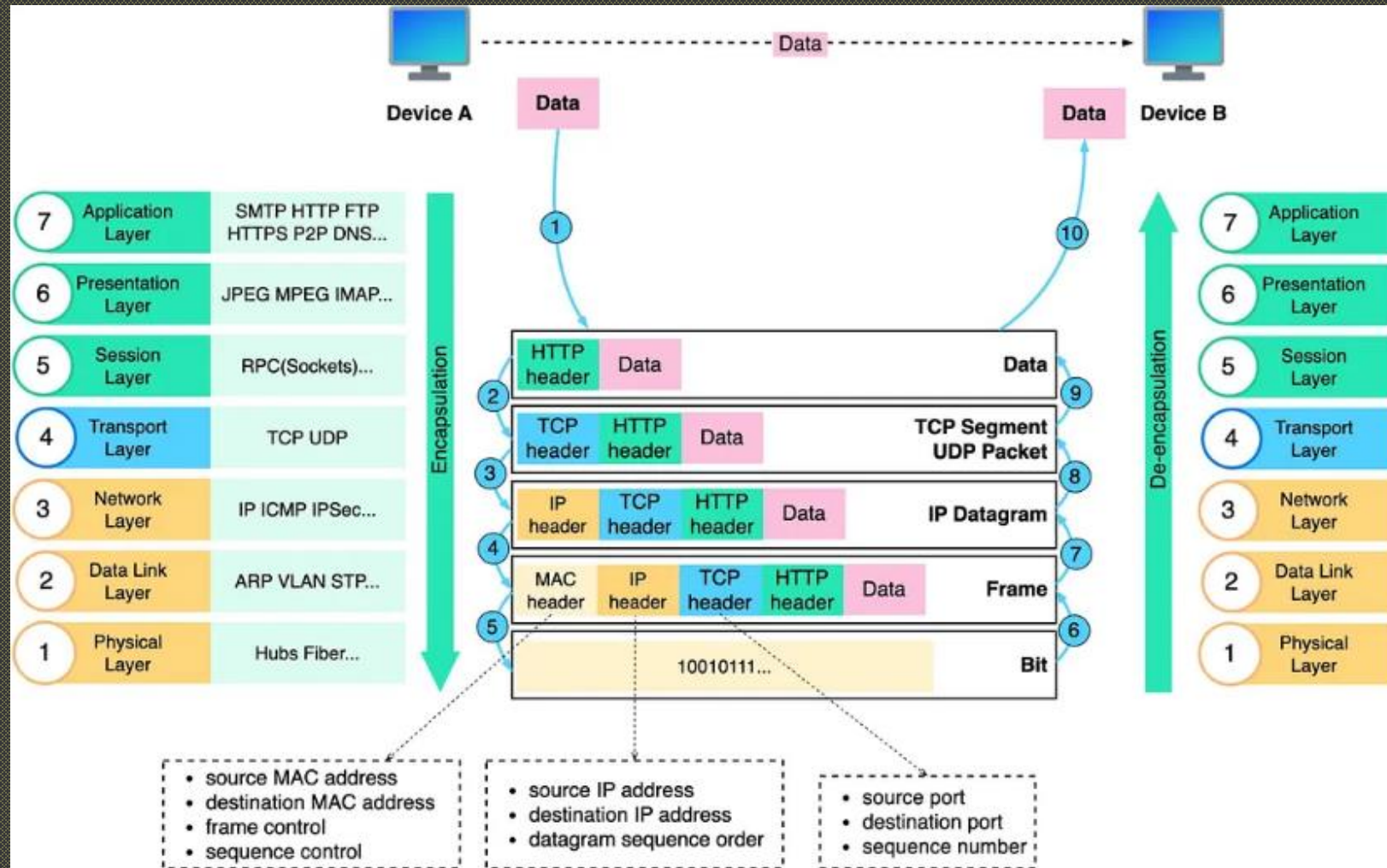


Generation	Speed	Example Use
Standard Ethernet	10 Mbps	Early office networks (1980s)
Fast Ethernet	100 Mbps	Basic modern LANs
Gigabit Ethernet	1 Gbps	Most current office/home LANs
10 Gigabit Ethernet	10 Gbps	Data centers, high-speed backbones

6.1 Ethernet



6.1 Ethernet



6.1 Ethernet

Standard Ethernet

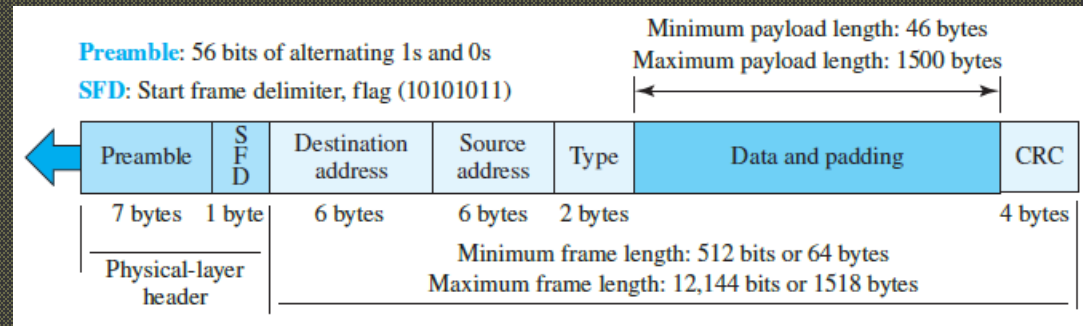
- The first version of Ethernet with a speed of 10 Megabits per second (Mbps).
- Although faster versions exist now, many core features from Standard Ethernet are still used today.
- It laid the foundation for Fast, Gigabit, and 10-Gigabit Ethernet.

In Standard Ethernet (10 Mbps):

- Minimum frame size = 512 bits.
- Transmission time for 512 bits = slot time = $51.2 \mu\text{s}$.
- Round-trip distance = $(2 \times 10^8) \times (51.2 \times 10^{-6}) = 10240 \text{ m}$
- One way of $\sim 5000 \text{ m}$; IEEE 802.3 caps total cable at 2500 m

Type of Service: Connectionless and Unreliable

- Connectionless:
 - Frames (data packets) are sent independently, not as part of a continuous connection.
 - No setup or termination of connection before or after sending data.
 - Example: Sending letters without calling the post office first - each letter travels on its own.
- Unreliable:
 - If a frame is lost or corrupted, Ethernet does not resend it.
 - Higher layers (like TCP) handle error checking and retransmission.
 - Example: If one of the messages gets lost, Ethernet will not fix it - TCP will handle it later.



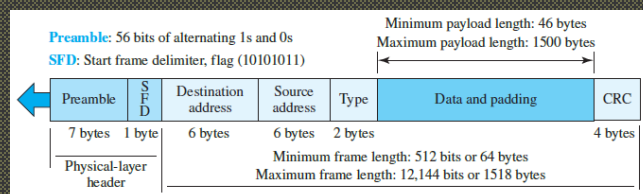
6.1 Ethernet

Standard Ethernet

Field	Size	Purpose / Explanation	Example / Note
1. Preamble	7 bytes	Alternating 0s and 1s to alert the receiver and synchronize timing.	Not part of frame data (added at physical layer).
2. Start Frame Delimiter (SFD)	1 byte (10101011)	Marks start of frame; warns receiver to get ready.	Added at physical layer.
3. Destination Address (DA)	6 bytes	Address of receiver (unicast, multicast, or broadcast).	e.g., FF:FF:FF:FF:FF:FF = broadcast to all.
4. Source Address (SA)	6 bytes	Address of sender.	e.g., your PC's network card.
5. Type	2 bytes	Identifies upper-layer protocol (like IP, ARP, OSPF).	e.g., 0800 = IPv4.
6. Data (Payload)	46–1500 bytes	Actual data from upper layers.	Padded if <46 bytes; split if >1500 bytes.
7. CRC (Cyclic Redundancy Check)	4 bytes	Used for error detection. If error found → frame is dropped.	CRC-32 value calculated by both sender & receiver.

The Ethernet frame arrives at the receiver's Network Interface Card (NIC). The NIC reads the Type field value. Indicates the higher-layer protocol encapsulated in the data field (e.g., 0x0800 for IPv4, 0x0806 for ARP). Values above 0x0600 denote the EtherType; below that, it is length (in older 802.3 frames).

Example: If Type = 0x0800, → pass to IPv4. If Type = 0x0806, → pass to ARP. If Type = 0x86DD, → pass to IPv6.



Type	Length (Bytes)	Explanation
Minimum Frame Length	64 bytes	Needed for CSMA/CD (collision detection).
Maximum Frame Length	1518 bytes	Prevents one device from using the network too long.
Minimum Data Length	46 bytes	If smaller, padding is added.
Maximum Data Length	1500 bytes	If larger, fragmentation occurs.

6.1 Ethernet

Standard Ethernet

Ethernet transmits bits differently from how we write them:

- Data is sent left to right, byte by byte.
- Inside each byte, the least significant bit (LSB) is sent first.
- This lets the receiver quickly identify if the frame is unicast (for one device) or multicast (for a group).
- Example: If the first bit received = 1 → multicast; if 0 → unicast.

Broadcast Address

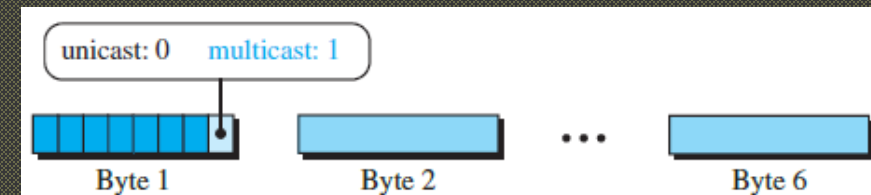
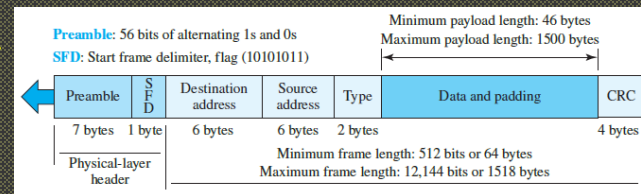
- Special type of Multicast → sent to all devices on the LAN.
- Every station receives and processes it.
- Used for network-wide announcements or device discovery.
- Example:
 - When a device first joins the network, it sends a broadcast to find the gateway (using ARP).
- Identification Rule:
 - Broadcast address = 48 bits of all 1s.
 - In hex → FF:FF:FF:FF:FF:FF.

Unicast Address

- Frame is sent to one specific device on the network.
- Only one NIC (Network Interface Card) receives and processes it.
- Example:
 - Your PC → sends data directly to one printer.
- Identification Rule:
 - If the least significant bit (LSB) of the first byte = 0, → it's a Unicast address.
 - Example MAC (hex): 4A:30:10:21:10:1A → first byte 4A (0100 1010 starts with 0).

Multicast Address

- Frame is sent to a group of devices (not everyone).
- Used for communication where multiple devices need the same data.
- Example:
 - Sending a video stream to multiple computers in a classroom.
- Identification Rule:
 - If the LSB of the first byte = 1, → it is a Multicast address.
 - Example MAC (hex): 47:20:1B:2E:08:EE → first byte 47 (0100 0111 starts with 1).



Type	Sent To	LSB of 1st Byte	Example MAC	Description
Unicast	One device	0	4A:30:10:21:10:1A	One-to-one communication
Multicast	A group of devices	1	47:20:1B:2E:08:EE	One-to-many communication
Broadcast	All devices	All 1s	FF:FF:FF:FF:FF:FF	One-to-all communication

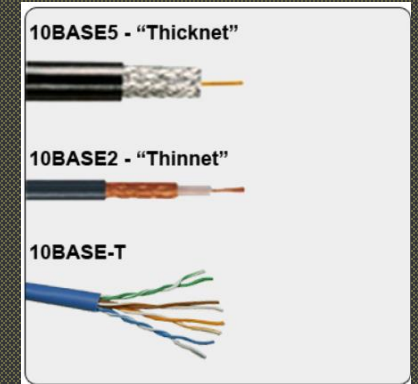
6.1 Ethernet

Standard Ethernet

- Different physical versions (cables and setups) of Standard Ethernet (10 Mbps).
- Each version used a different type of cable, length limit, and encoding method.
- Four main types became popular in the 1980s.

Each part of the name gives information about the Ethernet type:

- 10 → Data rate = 10 Mbps.
- Base → Baseband transmission (digital signal, not modulated).
 - Means bits are directly sent as digital pulses (not analog).
- X → Type or length of cable:
 - 5 → 500 m (thick coax)
 - 2 → 185 m (thin coax)
 - T → Twisted-pair cable (UTP)
 - F → Fiber-optic cable
- Baseband = only one signal (digital) is transmitted on the medium at a time.
- The signal represents bits (1s and 0s) directly - no modulation to different frequencies.
- Example: A 1 might be represented by a high voltage, and 0 by low voltage.
- Encoding (Sender side):
 - Data (bits) → converted into a digital line code before transmission.
 - This ensures synchronization and signal integrity.
- Decoding (Receiver side):
 - The incoming digital signal → interpreted and converted back to original bits.
- All implementations use line coding (process of converting digital data (bits of 1s and 0s) into a digital signal (voltages or light pulses) for reliable data transfer at 10 Mbps.



<https://www.itprc.com/overview-of-ethernet-history/>

Implementation	Cable Type (Medium)	Max Cable Length	Notes / Examples
10Base5	Thick coaxial cable	500 m	Called "Thick Ethernet"; used in early backbone networks.
10Base2	Thin coaxial cable	185 m	Called "Thin Ethernet"; cheaper and easier to install.
10Base-T	Twisted pair (UTP)	100 m	Most common in offices; uses RJ-45 connectors.
10Base-F	Fiber optic cable	2000 m	Used for long-distance connections; immune to electrical noise.

6.1 Ethernet

Fast Ethernet

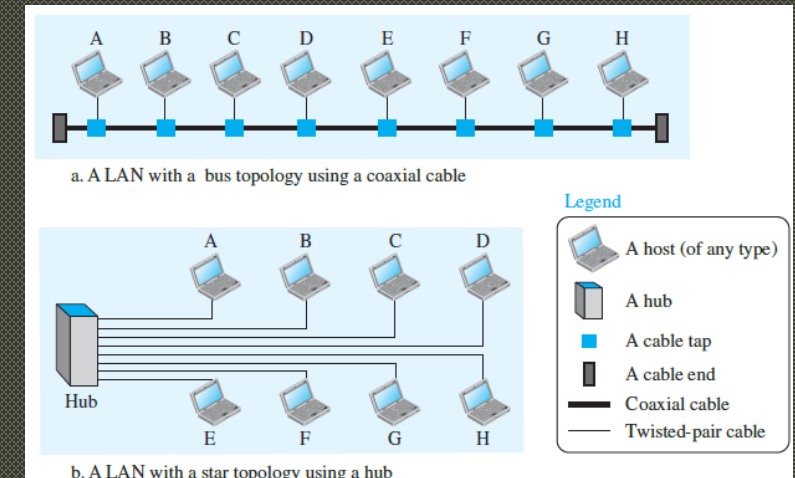
- Developed in the 1990s to compete with faster LAN technologies like FDDI.
- Increased Ethernet speed from 10 Mbps → 100 Mbps (10× faster).
- Still compatible with older (Standard Ethernet) systems.

Goals of Fast Ethernet

- Increase data rate to 100 Mbps.
- Remain compatible with Standard Ethernet.
- Keep the same 48-bit MAC address.
- Keep the same frame format.
 - Result: Only physical layer and access method were modified - not the MAC layer.

Auto-negotiation (New Feature)

- Allows devices to automatically agree on the best speed and mode.
- Useful when connecting devices with different capabilities (e.g., 10 Mbps ↔ 100 Mbps).
- Also lets a device check a hub or switch's speed capability.
- Example: A 10 Mbps printer connects to a 100 Mbps switch → both negotiate to operate at 10 Mbps.



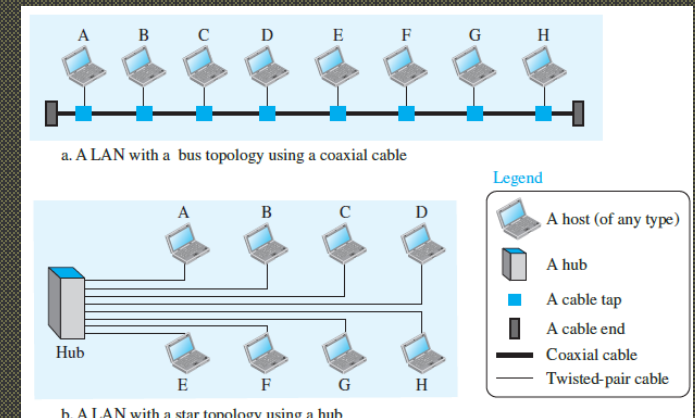
In Fast Ethernet (100 Mbps):

- Minimum frame size = 512 bits.
- Transmission time for 512 bits = slot time = 5.12 μ s.
- Round-trip distance = $(2 \times 10^8) \times (5.12 \times 10^{-6}) = 1024$ m
- One way of ~500 m; IEEE 802.3 caps total cable at 250 m

6.1 Ethernet

Fast Ethernet

- Ethernet uses CSMA/CD access method. Adjustments have been made for fast Ethernet.
- At higher speed (100 Mbps), collisions happen faster, so network length must shrink to detect them in time.
- Two Solutions:
 - Star Topology with Hubs
 - Replaces the long bus (max 2500 m → now 250 m).
 - Keeps backward compatibility with old Ethernet.
 - Switches + Full-Duplex Links
 - Each host has a dedicated link → no collisions.
 - No need for CSMA/CD (each link is private).
 - Switch stores frames in buffers and forwards them to the correct port.
- Example:
 - Old Ethernet: Shared cable, all devices compete.
 - Fast Ethernet with switch: Each PC has its own private line → faster and collision-free.



Encoding Methods

- Standard Ethernet used Manchester Encoding, but at 100 Mbps it needs 200 MHz bandwidth, too high for twisted-pair cables.
- Fast Ethernet used new encoding schemes depending on the cable type.

6.1 Ethernet

Fast Ethernet

Bit	What happens	Signal result
1	no change	stays low
0	change	goes high
0	change	goes low
1	no change	stays low
1	no change	stays low
0	change	goes high

NRZI Bits: 1 0 0 1 1 0

Encoding Explanation

- 4B/5B: Converts every 4 bits of data → 5 bits for synchronization (avoids long 0s or 1s).
- MLT-3 (Multi-Level Transmission 3): Uses 3 voltage levels to reduce signal frequency → good for twisted-pair.
- NRZ-I (Non-Return to Zero Inverted): Used in fiber systems; a signal change means a “1”.
- 8B/6T: Converts 8 bits → 6 ternary symbols (three voltage levels). The transmitter uses the lookup table to translate the binary data into ternary form.

Example:

- 100Base-TX sends data at 125 Mbps (after 4B/5B coding) → but because of MLT-3, the signal bandwidth is much lower, making it efficient for copper cables.

Implementation	Medium	Max Length	Wires	Encoding	Description
100Base-TX	Shielded or Cat 5 UTP	100 m	2 pairs	4B/5B + MLT-3	Common in LANs; efficient for twisted-pair cable.
100Base-FX	Fiber-optic	185 m	2 pairs	4B/5B + NRZ-I	Used for long distances; high reliability.
100Base-T4	Unshielded Twisted Pair (UTP Cat 3)	100 m	4 pairs	8B/6T	Used when Cat 5 cables are not available.

MLT-3 (Multi-Level Transmission – 3 levels):

- Uses three voltage levels: +1, 0, -1.
- “1” bit → signal moves to next level.
- “0” bit → signal stays at same level.
- Signal cycles: +1 → 0 → -1 → 0 → +1 ...

Why used in Fast Ethernet (100Base-TX):

- High-speed (100 Mbps) signals cause high frequency, leading to interference and loss on twisted-pair cables.
- MLT-3 lowers signal frequency by changing voltage less often.

How it reduces frequency:

- Takes 4 bits to complete one full voltage cycle.
- Signal frequency = $\frac{1}{4}$ of bit rate.
- Example: 125 Mbps data rate → 31.25 MHz signal frequency.¹³
 - Result: Lower frequency = less interference and signal loss → ideal for twisted-pair cables.

6.1 Ethernet

Gigabit Ethernet

Gigabit Ethernet (1000 Mbps, IEEE 802.3z)

- Main Goals
 - Increase data rate to 1 Gbps (1000 Mbps).
 - Stay compatible with Standard (10 Mbps) and Fast Ethernet (100 Mbps).
 - Keep the same 48-bit address.
 - Keep the same frame format.
 - Maintain the same min (64 bytes) and max (1518 bytes) frame sizes.
 - Support auto-negotiation (auto speed detection and setup).
- *Example:*

A computer using Fast Ethernet (100 Mbps) can upgrade to Gigabit Ethernet using the same frame structure - only the speed changes.
- MAC Sublayer
 - Keep the MAC layer mostly unchanged.
 - To achieve 1 Gbps, two access modes are defined:
 - Full-duplex mode (common)
 - Half-duplex mode (rare)

In Gigabit Ethernet (1000 Mbps):

- Minimum frame size = 512 bits.
- Transmission time for 512 bits = slot time = $0.512 \mu\text{s}$.
- Round-trip distance = $(2 \times 10^8) \times (0.512 \times 10^{-6}) = 102.4 \text{ m}$
- One way of $\sim 50 \text{ m}$; IEEE 802.3 caps total cable at 25 m

6.1 Ethernet

Gigabit Ethernet

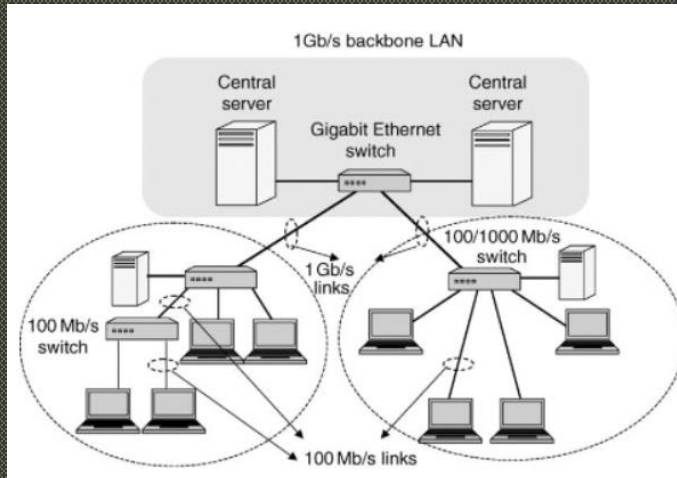
Full-Duplex Mode (Most Common)

- Uses a central switch connecting all devices.
- Each port has a buffer to store frames before sending.
- No collisions occur → no CSMA/CD needed.
- Max cable length depends on signal attenuation, not collision detection.
- Enables simultaneous send and receive.

• *Example:*

PC → Switch → Server

Each link is independent, so no collision can happen.



Half-Duplex Mode (Rare)

- Uses a hub (shared medium, like old Ethernet).
- Collisions possible, so CSMA/CD is used.
- Network length depends on minimum frame size and collision detection time.

Three methods handle this:

a. Traditional

- Keeps old minimum frame = 512 bits.
- But since data is 100× faster → collision detected too early.
- Slot time = $0.512 \mu\text{s}$ → max cable length $\approx 25 \text{ m}$ (too short).
 - *Example:* Only practical if all computers are in one room.

b. Carrier Extension

- Increases minimum frame to 512 bytes (4096 bits).
- Adds padding bits to small frames.
- Increases network length to 200 m (100 m hub-to-station).
 - *Example:* Adds dummy bits to make network length practical.

c. Frame Bursting

- For many short frames → sends them back-to-back instead of padding each one.
- Adds small gaps (padding) between frames to look like one long frame.
 - *Example:* Sends multiple small packets as one continuous burst to avoid idle time.

6.1 Ethernet

Gigabit Ethernet

Encoding

- Manchester encoding (used in 10 Mbps Ethernet) not suitable - too high bandwidth (2 Gbaud).
- 8B/10B encoding used for fiber and STP versions:
 - Converts 8 bits → 10-bit symbols for synchronization.
 - Works with NRZ (Non-Return-to-Zero) signaling.
- 1000Base-T uses 4D-PAM5 encoding:
 - 4 pairs transmit simultaneously.
 - Each symbol represents 5 voltage levels.

Imagine sending the 8-bit data 10110011 (that's "179" in decimal, like part of a file).

With 4D-PAM5:

- Map the 8 bits to a 4D symbol via encoding tables. Suppose it picks levels:
- Pair 1: +2V (codes "11")
- Pair 2: 0V (codes "00")
- Pair 3: -1V (codes "10")
- Pair 4: +1V (codes "01") + parity tweak to -2V for error check.
- Real mapping uses lookup tables, but this example might represent "11 00 10 01" = 10110011, with the tweak ensuring the total "sums" correctly for error detection.
- -2V is not directly from normal bits.
 - It is a parity/error check tweak applied to one of the four symbols in the 4D group.

• Example:

For 1000Base-T, each twisted pair carries $250 \text{ Mbps} \times 4 \text{ pairs} = 1 \text{ Gbps}$ total.

Implementation	Medium	Type	Max Length	Wires	Encoding
1000Base-SX	Fiber	Short-wave (S-W)	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber	Long-wave (L-W)	5000 m	2	8B/10B + NRZ
1000Base-CX	Shielded Twisted Pair (STP)	Copper	25 m	2	8B/10B + NRZ
1000Base-T	Unshielded Twisted Pair (UTP Cat 5)**	Copper	100 m	4	4D-PAM5

6.1 Ethernet

10 Gigabit Ethernet

- Increase data rate to 10 Gbps.
- Keep the same frame size and format as previous Ethernet standards.
- Allow interconnection of LANs, MANs, and WANs.
- Works over wired connections, often involving fiber optic cables, but also copper cabling.
- *Example:* Connecting two city buildings with fiber at 10 Gbps for very fast data transfer.

Physical Layers

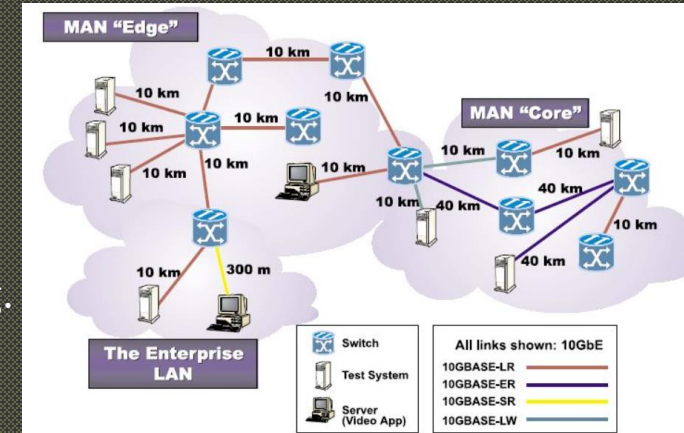
- LAN PHY: Designed to support existing LANs; links can connect through 64B/66B encoding.
- WAN PHY: Designed for wide-area networks; links can connect through 10GE.

Operation Mode

- Full-duplex only → no collisions → CSMA/CD not used.
- Each connection is dedicated, allowing simultaneous send and receive.
- *Example:* Two switches connected by 10GBase-LR fiber can exchange data at full 10 Gbps both ways at the same time.

Example:

- 10GBase-SR → short-distance fiber within a building.
- 10GBase-LR → connects different buildings in a campus or city.
- 10GBase-EW → connects cities over long distances.
- 10GBase-X4 → older multi-fiber implementation for medium distances.



<https://www.fiberoptics4sale.com/blogs/archive-posts/95046598-10-gigabit-ethernet-technology-and-lan-man-and-wan-networks?>

6.2 Wi-Fi

IEEE 802.11

- IEEE 802.11 specifies wireless LAN (WLAN) standards for the physical and data-link layers.
- WIFI sometimes called wireless Ethernet.

Architecture of Wireless LANs - two main service sets: BSS and ESS.

1. Basic Service Set (BSS): Building block of a WLAN.

- Components: Wireless stations (devices) + optional Access Point (AP). Types:

- Infrastructure BSS:

- Has an AP. Example: Your home Wi-Fi where your laptop connects to a router (AP).

- Ad hoc BSS:

- No AP; stations communicate directly. Example: Two laptops connecting via Wi-Fi without a router.

2. Extended Service Set (ESS): Multiple BSSs connected via a distribution system (wired or wireless).

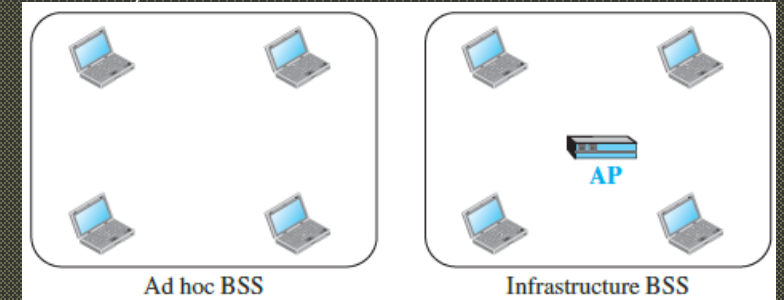
- Expands network coverage beyond one BSS.

- Stations:

- Mobile stations: Normal devices moving inside a BSS.
 - Stationary stations: APs connected to a wired LAN.

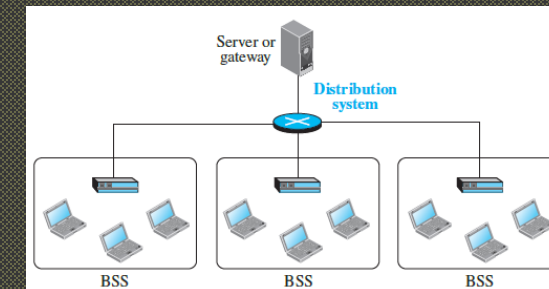
- Example:

- Humber campus Wi-Fi: multiple access points (BSSs) connected to the campus Ethernet.
 - Mobile devices can move between APs without losing connection to the network.



Station Types (Mobility)

- No-transition: Device stays inside one BSS (no movement).
 - Example: Desktop computer.
- BSS-transition: Device moves between BSSs inside one ESS.
 - Example: Student walking across campus staying connected.
- ESS-transition: Device moves between different ESS networks.
 - Example: Moving from campus Wi-Fi to home Wi-Fi.



6.2 Wi-Fi

IEEE 802.11

MAC Sublayer (How Wi-Fi Shares the Air). Wi-Fi uses two main methods to manage who can talk (send data):

a. Distributed Coordination Function (DCF)

- Most common method - used in all Wi-Fi networks.
- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).

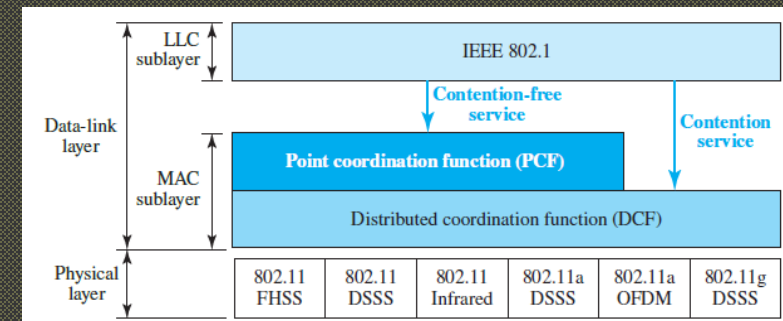
How it works:

- Device listens if the channel is free.
- If free → waits a short time (DIFS) → sends RTS (Request to Send).
- Receiver sends CTS (Clear to Send) → means “I am ready.”
- Device sends data → Receiver sends ACK (Acknowledgment).
- Example: Phone A wants to send a photo to laptop B → waits for silence → asks to send (RTS) → laptop replies yes (CTS) → photo sent → laptop says got it (ACK).

DIFS (DCF Interframe Space) is a fixed wait time for stations after the wireless medium idles before transmitting data frames.

b. Point Coordination Function (PCF)

- Optional (used in networks with an AP).
- The Access Point controls who speaks - avoids collisions.
- Used for real-time data (like video calls).
- Uses a polling system - AP asks devices one by one if they have data to send.

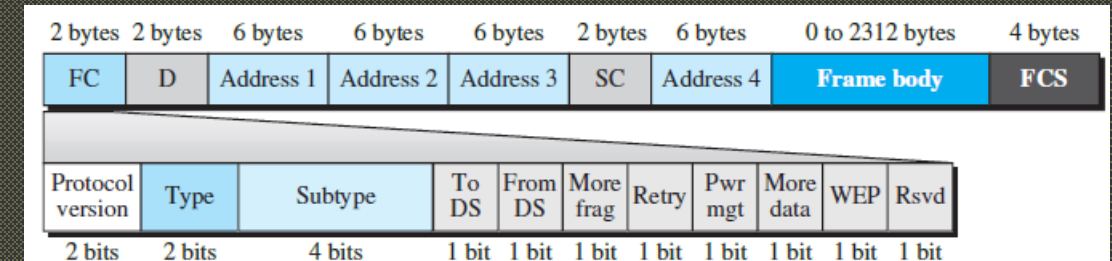


6.2 Wi-Fi

IEEE 802.11

Frames (Wi-Fi Messages)

- Wi-Fi sends data in frames (like envelopes).
- Each frame has:
 - Header info (type, source, destination).
 - Data (the actual message).
 - FCS (error check).
- Three types of frames:
 - Management: Connect/disconnect (like login/logout).
 - Control: For RTS, CTS, ACK.
 - Data: Actual user information (files, videos, etc.).



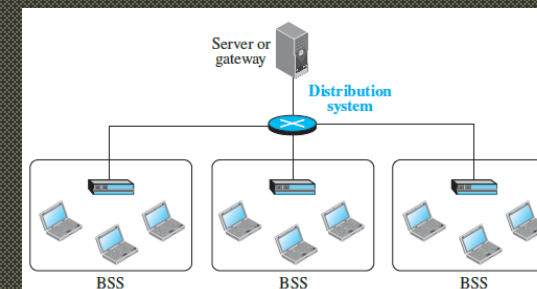
Standard	Frequency Band	Max Speed	Notes
802.11	2.4 GHz	1–2 Mbps	First version
802.11a	5 GHz	6–54 Mbps	Faster, shorter range
802.11b	2.4 GHz	5.5–11 Mbps	Popular, backward compatible
802.11g	2.4 GHz	22–54 Mbps	Backward with 802.11b
802.11n	2.4/5 GHz	Up to 600 Mbps	Uses multiple antennas

Addressing (How Wi-Fi Knows Where to Send Data)

- Wi-Fi frames can have up to 4 addresses depending on direction:
- Example: Your laptop (device) sends data to your router (To DS = 1, From DS = 0).

DS - Distributed System is the “backbone” that connects multiple Access Points (APs) together.

To DS	From DS	Example Meaning
0	0	Direct device-to-device (Ad hoc)
0	1	From router to device
1	0	From device to router
1	1	Between two routers (wireless backbone)



6.2 Wi-Fi

Wireless Access Point

- A Wireless Access Point (AP) is a central device that connects wireless computers (laptops, phones, tablets) so they can communicate with each other and access the network or Internet.
- Works like a hub, but instead of using cables, it uses radio waves (wireless signals).
- Found everywhere - schools, offices, airports, libraries, cafés, etc.

Central Communication Point

- All wireless devices (stations) send data through the AP.
- The AP receives signals and retransmits them to the destination device - just like a hub repeats signals.
- All devices can “hear” all the transmissions.
- *Example:*
If your laptop sends a file to a printer wirelessly, the data first goes to the AP → then the AP sends it to the printer.

Acknowledgments (ACK)

- After receiving data, the destination device sends back an ACK (acknowledgment) to confirm it got the message.
- The sender waits for this acknowledgment - if none is received, it resends the data.
- This prevents errors and collisions (two signals interfering).
- *Example:*
Like saying: “I got your message!” - to confirm you heard someone before they continue talking.



A router creates and manages the network, while an access point only extends or provides wireless access to it.

Device	Description	Used In
Wireless Router	Combines AP + Switch + Router in one box. Has antennas. Connects to a modem for Internet access.	Homes & small businesses
Access Point (AP)	Only provides wireless connection. Connects to a wired network or switch.	Large organizations, schools, enterprise networks

6.2 Wi-Fi

Wireless Access Point

- Shared Bandwidth
 - Like hubs, wireless networks share bandwidth among all connected devices.
 - If one device uses a lot of data, others get less speed.
 - *Example:* If 10 people stream videos on a 54 Mbps Wi-Fi network, each might only get ~5 Mbps or less.
- Effective Bandwidth
 - Actual usable speed is about half of the rated (physical) bandwidth due to:
 - Acknowledgment packets (ACK)
 - RTS/CTS messages
 - Signal interference or retransmissions
 - *Example:* 54 Mbps Wi-Fi network
 - Shared among 10 devices → 5.4 Mbps each
 - After overhead → ≈ 2.7 Mbps effective speed per device
- Improving Wireless Speeds
 - Modern wireless technologies (e.g., Wi-Fi 5, Wi-Fi 6, Wi-Fi 7) have drastically improved:
 - Speeds up to 1-10 Gbps
 - Better error control and less interference
 - More devices supported simultaneously
 - *Example:* New Wi-Fi 6 routers at Humber can handle hundreds of students streaming or uploading files at once.

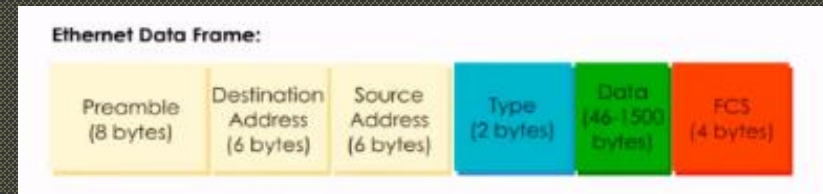


6.2 Wi-Fi

Difference between Ethernet and Wi-Fi

Both, 802.3 Ethernet and 802.11 Wi-Fi (or the latest 802.11ac) are TCP/IP networking technologies with differences in layer 1 & layer 2 of OSI model only:

- Ethernet for layer 1 of OSI model
 - Media could be UTP, fiber optics, and coaxial cable
 - Most popular topology is star
- Wi-Fi for layer 1 of OSI model
 - Media is air
 - Topology is star or fully meshed
- Ethernet for layer 2 of OSI model
 - Access method is CMA/CD
 - It uses layer 2 switch
 - Data frame header is shorter than Wi-Fi
- Wi-Fi for layer 2 of OSI model
 - Access method is CSMA/CA
 - It uses Access Point – a bridge that connects Ethernet to Wi-Fi
 - Data frame header is longer, which makes it less efficient than Ethernet



6.3 Bluetooth

Overview

- Bluetooth is a short-range wireless LAN technology.
- Used to connect nearby devices (phones, laptops, speakers, printers, etc.) without cables.
- Works only at short distances (about 10 meters).
- Commonly used for personal use - not for large networks.
- Belongs to IEEE 802.15 standard → called a Wireless Personal Area Network (WPAN).

How Bluetooth Works

- Bluetooth devices automatically find each other and form a small network.
- This small network is called a Piconet.
- If one device in the group has Internet access, it can share it with others.
- If too many devices try to connect → network becomes chaotic or slow.

Common Uses

- Wireless mouse, keyboard, or earbuds connected to a computer or phone.
- Smart home devices (door sensors, alarms, lights).
- Health monitoring (e.g., smartwatches sending data to a phone).
- File sharing between phones or laptops.

- Bluetooth = short-range, low-power device-to-device communication.
- Wi-Fi = longer-range, high-speed network communication.



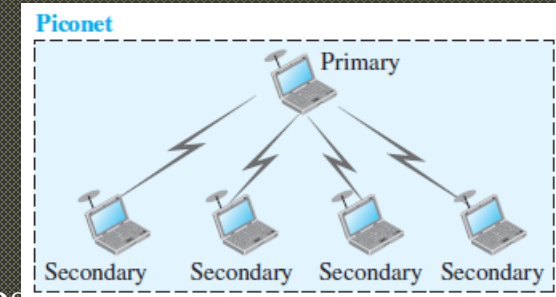
<https://www.mokoblue.com/bluetooth-vs-wifi-which-is-better/>

6.3 Bluetooth

Bluetooth Architecture

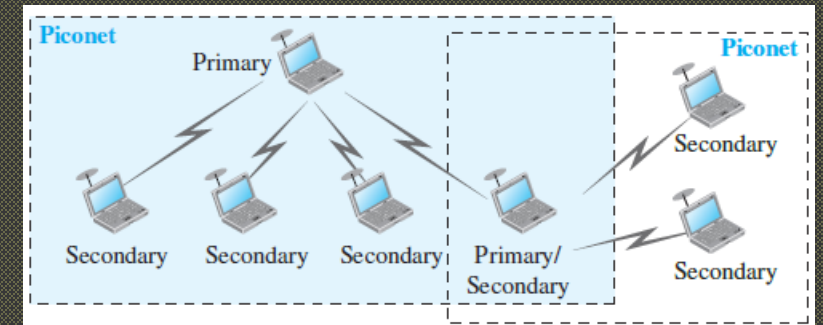
1. Piconet (small network)

- One small network with up to 8 active devices.
- 1 device = Primary (Master) → controls communication.
- Up to 7 devices = Secondary (Slaves) → follow the primary.
- Example:
 - Your phone (primary) connects to earbuds, smartwatch, and laptop (secondaries).
- Other devices can join in a parked state - connected but not active until needed.



2. Scatternet

- Combination of multiple piconets.
- A device can belong to two piconets (secondary in one, primary in another).
- Example:
 - Your phone is a secondary to your laptop, but a primary to your earbuds.
- Allows larger networks but still for short distances.



Bluetooth Devices

- Have a short-range radio transmitter.
- Speed: about 1 Mbps.
- Frequency band: 2.4 GHz (same as Wi-Fi).
- Can cause interference with Wi-Fi networks nearby.

6.3 Bluetooth

L2CAP Functions (Simplified)

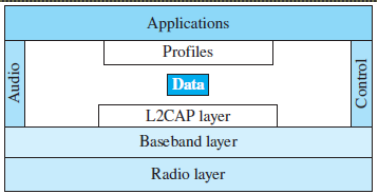
- Multiplexing: It takes messages from different apps and sends them all through one shared Bluetooth connection.
- Segmentation/Reassembly: Breaks big data into small packets, then rebuilds them.
- Group Management: Allows one device to send to a group (like a broadcast).
- QoS (Quality of Service): Ensures proper timing and priority for data.

Applications/Profiles are like rules or templates that tell devices how to use Bluetooth for a specific purpose.

Baseband Layer Details

- Works like a traffic control system - decides who can talk and when.
- Uses TDMA (Time Division Multiple Access) = time is split into slots (625 µs each).
- Devices take turns sending/receiving data.
- Primary talks in even slots → Secondary replies in odd slots.

Type	Meaning	Used For	Notes
SCO (Synchronous Connection-Oriented)	Real-time, no delay	Voice/ audio	No retransmission
ACL (Asynchronous Connection-Less)	Reliable, ensures no data loss	Files/ data	Retransmits if needed



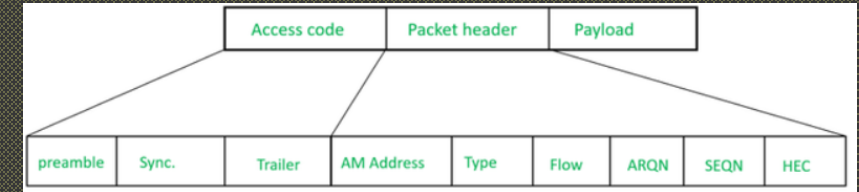
Layer	What It Does	Example
Radio Layer	Sends/receives signals using radio waves	Physical connection
Baseband Layer	Controls how devices share the air (timing, access)	Like MAC layer in LAN
L2CAP (Logical Link Control and Adaptation Protocol)	Handles data exchange, splitting, reassembling, and routing	Makes sure data fits Bluetooth packets
Applications/Profiles	Define what the connection is for	e.g., Audio, File Transfer, Mouse, Keyboard

6.3 Bluetooth

Frame Structure (Simplified)

Each Bluetooth frame has:

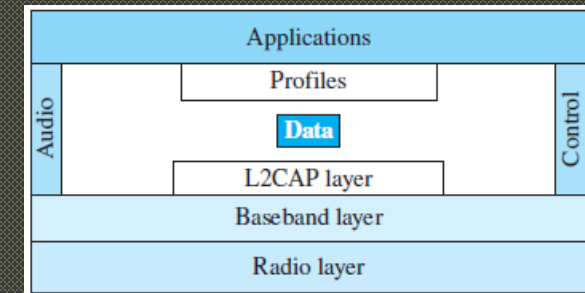
- Access Code (72 bits): Identifies the primary device.
- Header (54 bits): Contains address, type, and control bits.
- Payload: Actual data (up to ~2,740 bits).



<https://www.geeksforgeeks.org/computer-networks/bluetooth-frame-structure/>

Radio Layer

- Range: ~10 meters.
- Frequency Band: 2.4 GHz → 79 channels (1 MHz each).
- Uses FHSS (Frequency Hopping Spread Spectrum):
 - Changes frequency 1,600 times per second to avoid interference.
 - Each “hop” lasts 625 μs (microseconds).
- Modulation: GFSK (Gaussian Frequency Shift Keying).
 - Bit 1 = frequency goes up.
 - Bit 0 = frequency goes down.



FHSS - Frequency-Hopping Spread Spectrum

Concept	Simple Meaning	Example
Bluetooth	Short-range wireless tech	Connect phone to speaker
Piconet	Small group (1 main + 7 others)	Laptop with mouse + keyboard
Scatternet	Several piconets linked	Laptop connected to phone and printer
Range	~10 meters	Across a room
Speed	1 Mbps	Send a photo or song
Access Method	TDMA (takes turns)	Like walkie-talkies
FHSS	Hops frequencies fast	Reduces interference

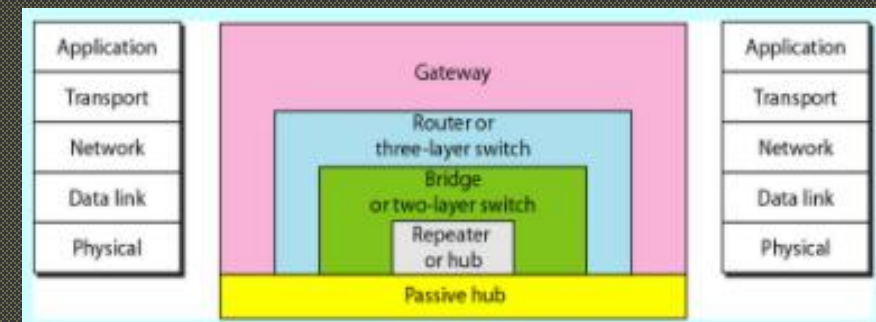
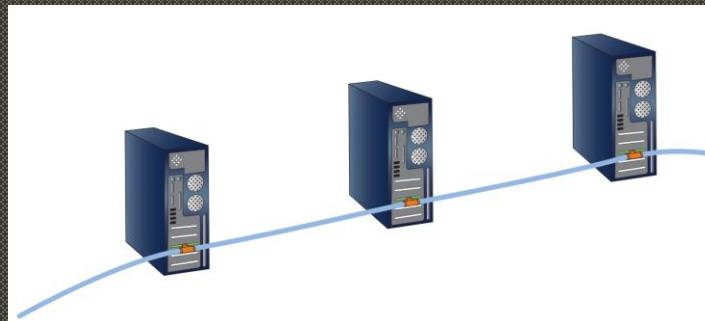
6.4 Repeaters & Hubs

Early Computer Networks

- Before interconnecting devices existed, computers were connected in a daisy-chain (series connection).
 - Example: Computer A → Computer B → Computer C → Computer D (all connected in one long cable).
- Problem:
 - The total cable length was limited - signals could only travel a certain distance before becoming too weak.
 - The number of computers that could be connected was limited because each device added resistance and signal loss.

The Problem with Long Cables

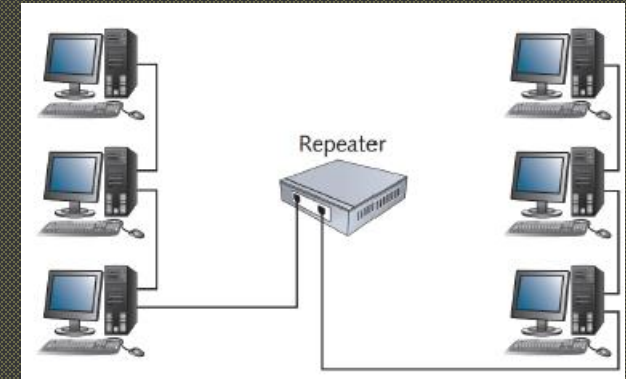
- Electrical signals weaken as they travel through a cable because of signal attenuation.
- When the signal becomes too weak, the receiving computer cannot understand the data.
- Example: If you shout down a long hallway, your voice becomes faint. Similarly, data signals fade as cable length increases.



6.4 Repeaters & Hubs

The Solution: The Repeater

- A repeater was invented to solve this problem.
- Function:
 - It receives the incoming data signal from one part of the network.
 - It cleans or regenerates the signal (not just amplifies noise).
 - It retransmits the signal at its original strength to the next segment of cable.
- Think of a repeater like a microphone and speaker system:
 - When a speaker's voice fades at the back of the room, a microphone picks it up and sends it through speakers so everyone can hear clearly.

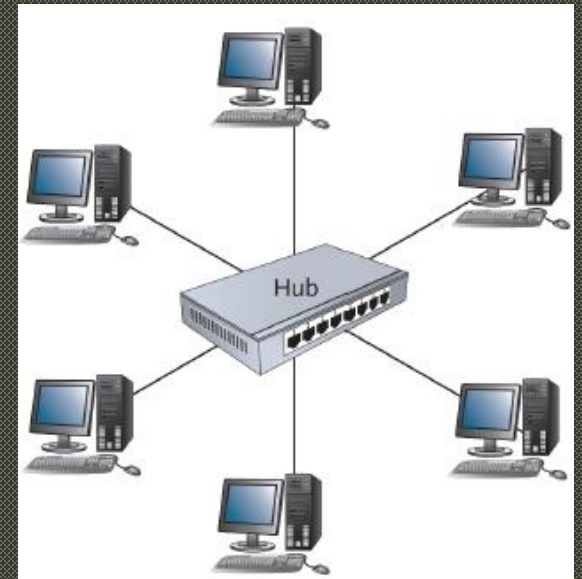


How a Repeater Works

- A traditional repeater has two ports (two connections):
 - Port 1: Receives the weak signal.
 - Port 2: Sends out the restored (strong) signal.
 - This allows you to extend the length of your network by connecting more cable segments.
 - Example:
 - Without a repeater: max distance between computers might be 100 meters.
 - With a repeater: two 100-meter cables can be connected for a total of 200 meters.
- Repeaters do not amplify the original electrical signal (they do not make it louder).
 - Instead, they recreate the signal perfectly at full strength - removing distortion and noise.
 - Example:
 - To connect computers or network segments that are too far apart for direct connection.
 - To maintain signal quality over long distances.

6.4 Multiport Repeaters & Hubs

- A multiport repeater is basically a repeater with more than two ports.
- It can have 4, 8, 12, or even 24 ports for connecting multiple computers or network devices.
- This makes it possible to connect many devices to a single central point, instead of chaining them one after another.
- A multiport repeater is commonly called a hub.
- The term “hub” is shorter and more widely used.
- So, in practice, when people say “hub”, they mean a multiport repeater.
- Network Design Changes
- Old way (daisy-chain):
 - Computer A → Computer B → Computer C (connected in a line).
 - Problem: if one computer or cable failed, the whole chain could be affected.
- New way (with hub):
 - All computers connect directly to one central hub using separate cables.
 - Looks like a star shape (called a star topology).
 - Example:
 - Port 1 → Computer A
 - Port 2 → Computer B
 - Port 3 → Computer C

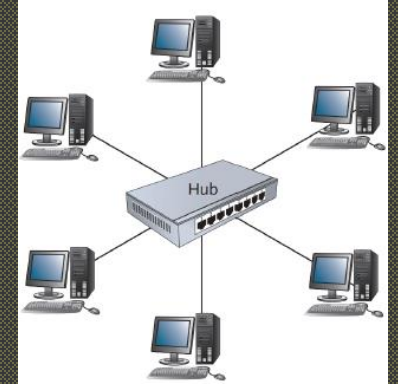


6.4 Multiport Repeaters & Hubs

A hub performs the same main job as a repeater, but for multiple connections.

Here's what it does step by step:

- Receives signals (data bits) from one of the computers connected to one of its ports.
- Cleans the signal: removes unwanted electrical noise or distortion.
- Regenerates the signal: restores it to its original strength and quality.
- Transmits (repeats) the regenerated signal to all other ports on the hub.
- Example:
 - If Computer A sends data, the hub receives the data and then sends that same data to Computers B, C, and D through its other ports.



The Hub broadcasts data to all connected devices (not just the intended one).

Power and Types of Hubs

- Hubs need electrical power to work because they must clean and regenerate signals.
- For that reason, they are sometimes called “active hubs.”
- There also exist “passive hubs”, which don't require power, but they don't regenerate the signal - they only act like simple connectors (and are rarely used today).

6.4 Multiport Repeaters & Hubs

- Bandwidth is the amount of data that can be sent or received on a network in one second.
- Unit of measurement: bits per second (bps).

Common speeds:

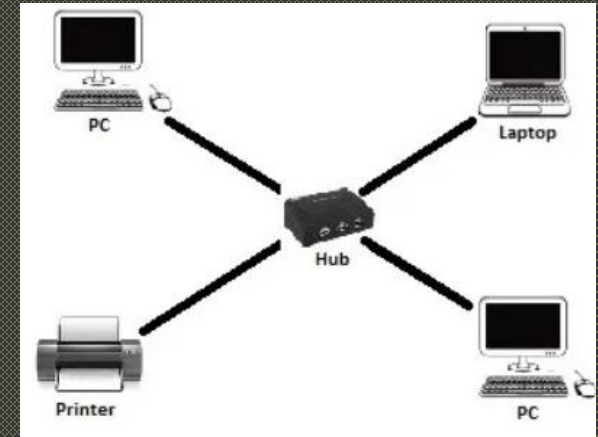
- 10 Mbps = 10 million bits per second
- 100 Mbps = 100 million bits per second
- 1 Gbps = 1 billion bits per second
- 100 Gbps = 100 billion bits per second

The Problem: Bandwidth Sharing in Hubs

- Hubs make all connected computers share the same total bandwidth.
- Only one computer can send data at a time through the hub.
 - If multiple computers try to send data at once, their signals collide, causing delays.
- This is called bandwidth sharing.

Example of Bandwidth Sharing

- A 10 Mbps hub with 10 connected computers → all share the same 10 Mbps.
- If all 10 computers are active at once:
 - Each one effectively gets about 1 Mbps ($10 \text{ Mbps} \div 10 \text{ computers}$).

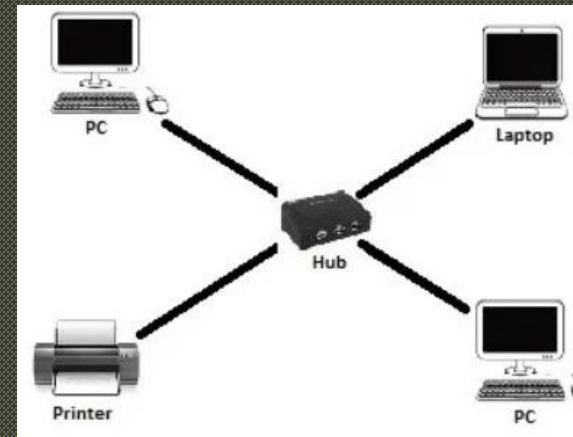


<https://medium.com/networks-security/collision-vs-broadcast-domain-hub-switch-and-router-5957331782d>

6.4 Multiport Repeaters & Hubs

- In the early days of LANs, hubs were fine because:
 - There were fewer computers.
 - Files were small (text, spreadsheets, small programs).
 - Network traffic was light (few people transferring large files at once).
- So, even though bandwidth was shared, networks still felt fast enough.
- Today's LANs transfer large multimedia files (e.g., HD videos, images, music, software).
- Many users access network applications and databases at the same time.
- Shared bandwidth creates severe slowdowns and congestion.
- Because of this, network administrators stopped using hubs in modern designs.
- Hubs have been replaced by switches, which give dedicated bandwidth to each device (so no sharing).
- As a result, it is now hard to find hubs for sale from major computer hardware stores.

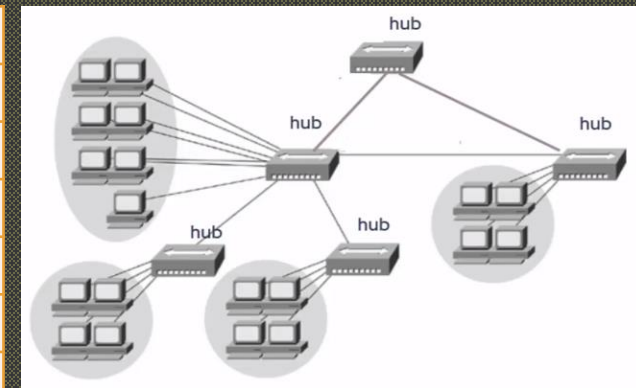
- Bandwidth = data transfer capacity of the network.
- Hubs share bandwidth → only one device can send data at a time.
- Their main disadvantage: shared bandwidth, which causes collisions and slow speeds.
- More devices = less speed per device.
- Switches replaced hubs because they provide full bandwidth per connection (no sharing).



6.4 Multiport Repeaters & Hubs

- Most network hubs have LED indicator lights on the front panel.
- These lights help network technicians quickly check the hub's status and activity without special tools.
 1. If the power light is off, the hub may be unplugged or broken.
 2. It stays off if no device is linked or the cable is defective
 3. A speed light shows the connection speed, like green for 100 Mbps.
 4. A blinking activity light shows data is being sent or received.
 5. A flashing collision light means devices are trying to talk at once.
 6. A utilization meter shows how much network bandwidth is being used.
 7. The uplink port and its button let you connect to other hubs or computers.

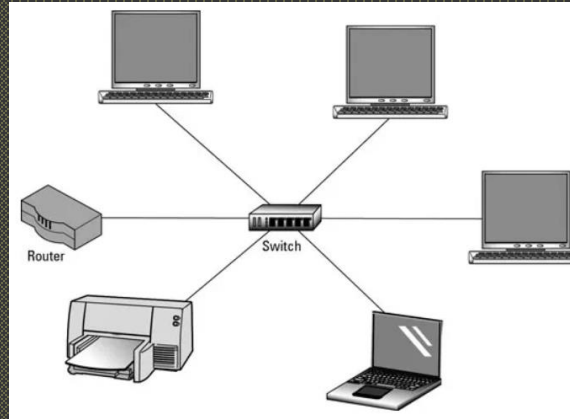
Indicator Type	Purpose	Typical Color/Behavior
1. Power	Shows if hub has power	Green (solid)
2. Link Status	Confirms valid cable connection	Green (solid)
3. Speed	Shows data rate (10/100 Mbps)	Amber = 10 Mbps, Green = 100 Mbps
4. Network Activity	Shows when data is transmitted	Blinking light
5. Collision	Warns of simultaneous transmissions	Flashing red or amber
6. Utilization	Shows total network traffic load	Multi-light bar
7. Uplink	Used to connect hub to another hub/switch	Button toggles mode



<https://medium.com/networks-security/collision-vs-broadcast-domain-hub-switch-and-router-5957331782d>

6.5 Network Switches

- A network switch is a device used to connect multiple computers or devices in a network so they can communicate.
- It looks like a hub - a box with multiple ports where you plug in network cables.
- But it is much smarter than a hub. The switch uses source and destination MAC addresses to know where to send the frame.
- A hub repeats signals to everyone (like shouting in a room).
- A switch sends data only to the intended station.



<https://medium.com/networks-security/collision-vs-broadcast-domain-hub-switch-and-router-5957331782d>

Feature	Hub	Switch
Function	Regenerates and sends data to all ports	Reads the data and sends it only to the correct port
Works With	Electrical signals (bits only)	Actual data (frames with addresses)
Efficiency	Wastes bandwidth (broadcasts to everyone)	Saves bandwidth (sends to the right device)
Collisions	Frequent	Rare or none
Performance	Slower in busy networks	Much faster and more efficient

6.5 Network Switches

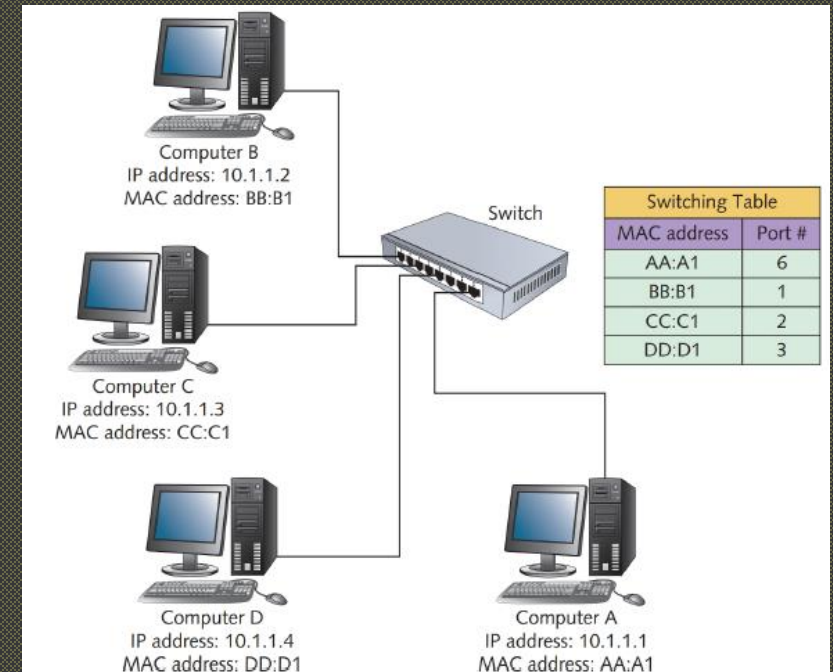
When a frame enters a switch, the switch performs the following actions:

- Receives the frame from one of its ports.
- Reads the source MAC address (who sent it) and the destination MAC address (who should receive it).
- Checks its switching table (a memory list of MAC addresses and their ports).
- Forwards the frame only to the correct port connected to the destination computer.
- Updates the switching table with new source information (if needed).

Scenario:

- PC B sends a frame to PC C.
 - Switch reads the source (PC B's MAC) and destination (PC C's MAC).
 - It learns "PC B is on Port 1" and saves it in its table.
 - It checks the table for PC C
 - If it finds PC C → sends frame only to Port 2.
 - If it does not find PC C → sends the frame to all ports (like a hub).

Computer	MAC Address	Port
PC1	AA:AA:AA:AA:AA:A1	Port 1
PC2	BB:BB:BB:BB:BB:B1	Port 2
PC3	CC:CC:CC:CC:CC:C1	Port 3
PC4	DD:DD:DD:DD:DD:D1	Port 4



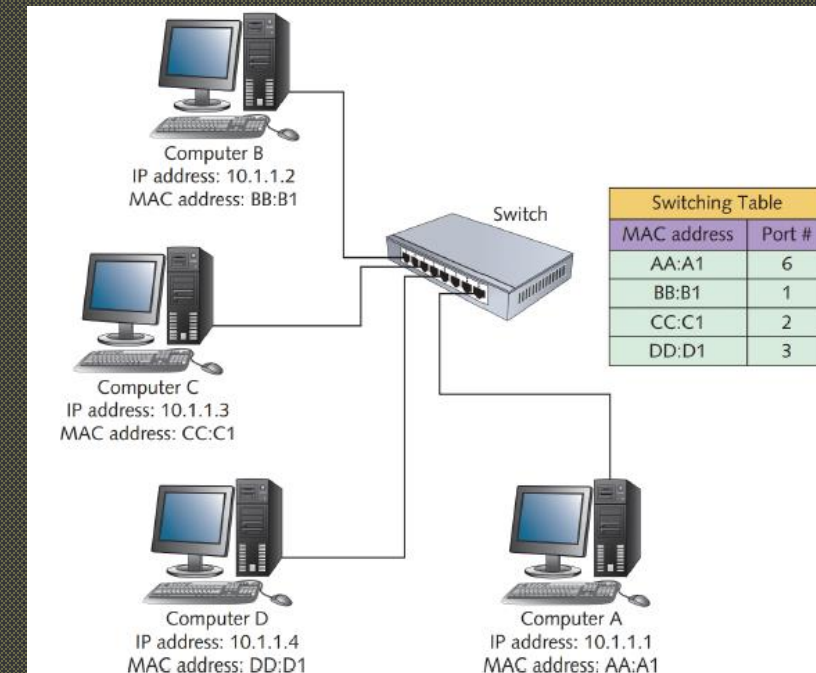
6.5 Network Switches

The Switching Table (MAC Address Table)

- The switch learns which device is on which port by reading the source address of each incoming frame.
- It stores this information in a switching table (also called the MAC address table).
- The timestamp shows when the entry was last updated.
- This table helps the switch know exactly where to send data next time.

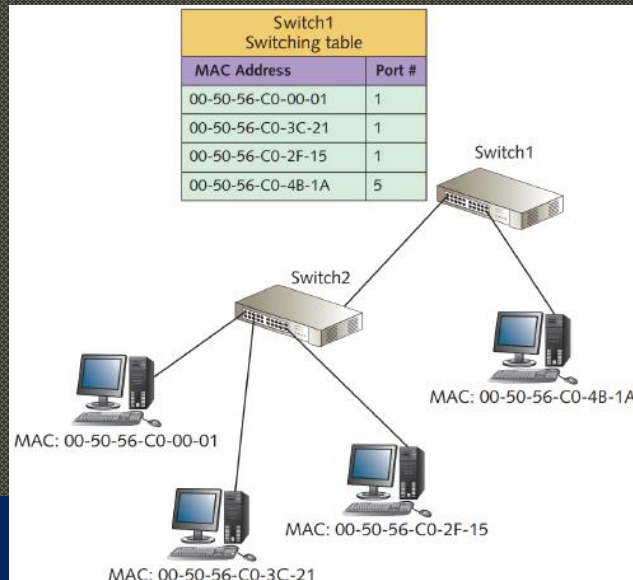
MAC Address	Port	Timestamp
BB:BB:BB:BB:BB:B1	1	18:00
CC:CC:CC:CC:CC:C1	2	18:01

- MAC addresses do not stay in the table forever - devices can move or disconnect.
- To avoid outdated information, each entry has a lifetime limit (often around 5').
- If the switch does not see traffic from that MAC address within the time limit:
 - The entry is deleted from the table.
- This ensures the switch table is accurate and current.
- Example:
 - If PC C is turned off for more than 5 minutes → its entry is removed.
 - When PC C reconnects and sends data, the switch learns it again and re-adds it.



6.5 Network Switches

- A switching table maps a device's MAC address to the switch port it's connected to.
- A switch starts with an empty table and learns device locations by examining traffic.
- The switch learns a MAC address from the source of a frame and records which port it came from.
- The switch forwards a frame to the port in its table for the destination MAC; if unknown, it floods the frame to all ports.
- A single port can have multiple MAC addresses if it's connected to another network device like a switch.
- Table entries have a timer and are automatically deleted if not refreshed by new traffic.
- A switching table contains entries like a MAC address, its associated port, and a timestamp.
- The switching table improves performance by sending traffic only to the correct port, reducing unnecessary flooding.



Concept	Description	Example
Switching Table	List of MAC addresses and ports	MAC A → Port 2
Learning	Switch adds source MACs to table	Learns PC1 on Port 2
Forwarding	Sends to known port	PC1 → PC2 via Port 5
Flooding	Sends to all ports when unknown	Broadcast to find new MAC
Multiple MACs per Port	Happens when other switches connect	Switch-to-switch link
Aging Time	Removes inactive entries	Deletes if no traffic for 5 min

6.5 Network Switches

- A switch sends data (frames) only to the destination device's port, not to every port like a hub does.
- This allows a switch to manage multiple simultaneous conversations between computers.
- Each connected device gets its own dedicated bandwidth, instead of sharing one common pool.

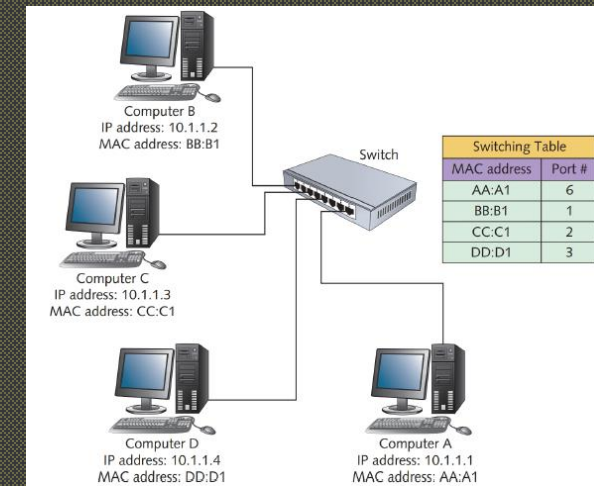
Example: Dedicated Bandwidth

- Imagine a 100 Mbps switch with 4 computers: A, B, C, and D.
- Hub:
 - All 4 computers share the same 100 Mbps.
 - If all are active, each may get only ~25 Mbps.
- Switch:
 - Computer A ↔ Computer C = 100 Mbps connection.
 - Computer B ↔ Computer D = 100 Mbps connection.
 - Both conversations happen at the same time, with no interference.
 - Each pair gets the full 100 Mbps speed.

- A switch allows simultaneous communication between multiple computers.
- Each device gets dedicated, full bandwidth (not shared).
- Full-duplex mode doubles communication speed (sending + receiving).
- Hubs = half-duplex, slower, more collisions.
- Switches = full-duplex, faster, collision-free.
- Because of these advantages, switches replaced hubs in all modern networks.

Example of Full-Duplex Operation

- If a computer is connected to a 100 Mbps switch port in full-duplex mode:
 - It can send data at 100 Mbps and receive data at 100 Mbps simultaneously.
 - So, total effective communication rate = 200 Mbps (100 Mbps each way).
- This is why switches are much faster and more efficient — no waiting, no collisions.

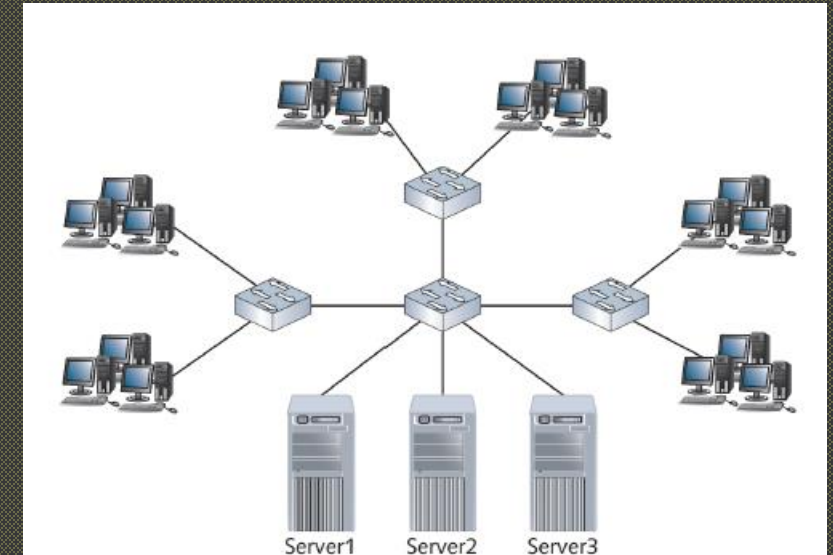


6.5 Network Switches

- Switches have LED lights (indicators) to help you quickly check the status and operation of each port.
- These lights make it easy to troubleshoot and monitor network activity without needing special tools.
 - The power light shows if the switch is on.
 - The link light shows a device is properly connected.
 - The activity light flashes when data is being sent or received.
 - The duplex light shows if a port can send and receive data at the same time.

Indicator Type	What It Shows	Common Label	Typical Light Behavior
Power	Switch is on/off	POWER	Solid green = on
Link Status	Device is connected	LINK / LNK	Solid green = link active
Activity	Data being sent/received	ACT	Blinking = network traffic
Duplex Mode	Full or half duplex	FDX / DUPLEX	On = full-duplex mode

- Uplinking = Connecting one switch to another to expand the network (add more devices).
- The cable used for this is called an uplink connection or trunk link.
- This allows your LAN to grow beyond the number of ports on a single switch.
- Many switches have a dedicated “Uplink” port - designed to connect directly to another switch.
- Some modern switches can use any regular port for uplinking by using an Ethernet crossover cable or auto-sensing port (Auto-MDI/MDIX).
- The uplink port sends and receives traffic between switches, keeping all devices in the same network.



6.5 Network Switches

1. Plug-and-Play Functionality

- Basic switches are *ready to use* right after being powered on.
- No configuration needed - just connect the devices, and the switch starts forwarding data (frames).

Example:

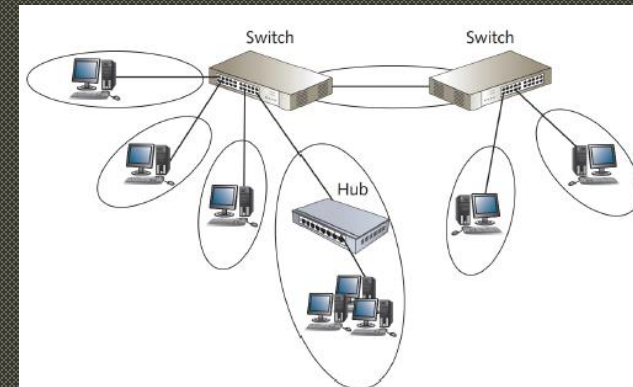
- Plugging in 4 computers into a simple 5-port switch.
- The switch automatically lets them communicate on the LAN - no setup required.

2. Layer of Operation

- Switches operate at the Data Link Layer (Layer 2) of the OSI Model.
- At this layer:
 - Devices use physical (MAC) addresses to identify themselves.
 - The Protocol Data Unit (PDU) is called a frame.
- Function:
 - The switch receives frames on one port and forwards them to the port where the destination device is connected.

3. Broadcast Frames and Flooding

- Broadcast frames are sent to all devices in a network.
- The destination address is FF:FF:FF:FF:FF:FF (all binary 1s).
- When a switch receives a broadcast frame:
 - It floods the frame - sends it out all ports except the one it came from.
 - Broadcasts are useful for tasks like discovering devices on the network.



Example:

- When a new computer joins a LAN, it sends a broadcast asking, “Who is the DHCP server?”
- The switch floods this message to all devices so the DHCP server can reply

6.5 Network Switches

4. Collision Domains

- Each switch port is its own collision domain.
- A collision domain is the area where data collisions can occur when two devices try to send data at the same time.
- If a collision occurs between a device and the switch port:
 - The switch does not forward the collision to other ports.
 - This isolates collisions, improving network performance.

Example:

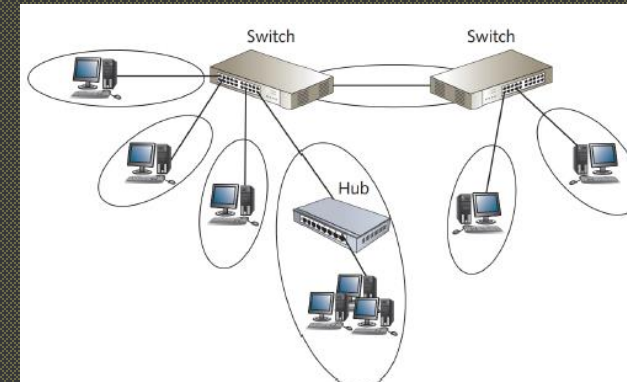
- If a printer and a PC share a hub and collide, that is one collision domain.
- But if each device connects to a switch port, their collisions are isolated - one device's issue does not affect others.

5. Full-Duplex Mode

- Full-duplex mode allows devices to send and receive data at the same time - no collisions possible.
- Both the switch port and the connected device (like a PC's NIC) must support full-duplex.
- Even though collisions cannot occur, each port is still referred to as a collision domain (for classification purposes).

Example:

- A full-duplex 100 Mbps connection can send 100 Mbps and receive 100 Mbps simultaneously.
- Effective total = 200 Mbps throughput.



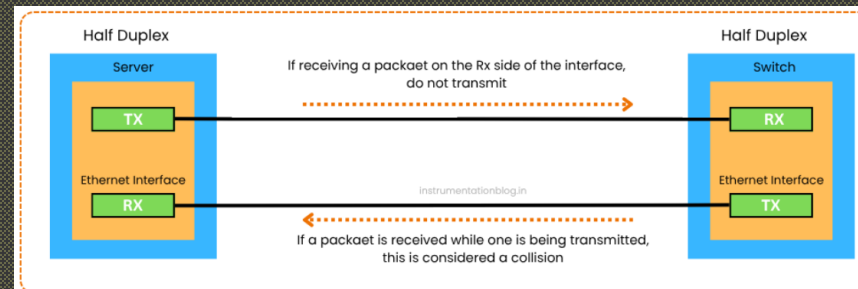
6.5 Network Switches

1. Switches Support Multiple Speeds and Duplex Modes

- Switch ports can operate at different speeds and different duplex settings depending on the connected device.
- Common speeds and duplex combinations for a 10/100/1000 Mbps switch include:
 - 10 Mbps half-duplex
 - 10 Mbps full-duplex
 - 100 Mbps half-duplex
 - 100 Mbps full-duplex
 - 1000 Mbps full-duplex
 - 1000 Mbps (1 Gbps) half-duplex mode does not exist in Ethernet (Gigabit Ethernet requires full-duplex operation).

2. Auto-Negotiation Mode

- Most modern switches use auto-negotiation, which means:
 - The switch and the connected device communicate automatically to determine the best possible speed and duplex mode both can support.
- This helps ensure maximum performance without manual configuration.
- Sometimes Auto-Negotiation Fails or,
 - If the switch supports 1 Gbps and a PC connected to it supports only 100 Mbps:
 - The switch automatically sets the port to 100 Mbps full-duplex to match the PC's capabilities.
- Occasionally, the negotiation between devices does not work correctly.
 - Result: The link may fail to connect or work at lower performance (e.g., mismatched duplex settings).
 - Fix: Manually configure both devices (switch port and NIC) to the same speed and duplex mode.



<https://www.instrumentationblog.in/auto-negotiation-in-networking/>

6.5 Network Switches

3. Manual Port Configuration (on Mid-Range or High-End Switches)

- Advanced switches let administrators manually configure each port for:
 - Speed (10, 100, or 1000 Mbps)
 - Duplex (half or full)
- This manual control ensures stability in environments where auto-negotiation can cause issues.

Example:

- In industrial or server environments, admins often lock ports to “1000 Mbps full-duplex” to prevent negotiation problems.

4. Auto-MDIX (Media-Dependent Interface Crossover)

- Auto-MDIX automatically detects the type of cable and device connected to a switch port.
- If needed, the switch automatically swaps its transmit (TX) and receive (RX) pins internally.
- This means it allows to use either a straight-through or crossover cable, and it will still work correctly.
- Without Auto-MDIX:
 - You would need to use a crossover cable when connecting two switches or two computers directly.
 - You would use a straight-through cable when connecting a PC to a switch.

Example:

- If you connect two switches using a regular straight-through Ethernet cable, Auto-MDIX automatically flips the transmit/receive pairs - no need for a special crossover cable.

6.5 Network Switches

5. Cost vs. Capability

- Basic (inexpensive) switches:
 - Use auto-negotiation and auto-MDIX.
 - Have limited manual configuration options.
- Advanced (enterprise) switches:
 - Allow per-port configuration (speed, duplex, auto-MDIX on/off).
 - Provide better performance, customization, and troubleshooting control.

Example:

- In a home or small office, a cheap plug-and-play switch works fine.
In an industrial network, you would invest in a managed switch that allows fine-tuned port settings.

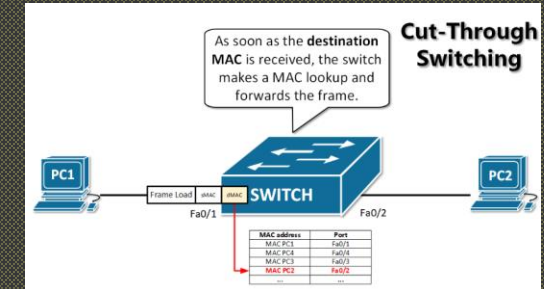
Feature	Description	Example
Speed Modes	10/100/1000 Mbps support	Adjusts to match device
Duplex Modes	Half (send or receive) or Full (both)	Full-duplex = better performance
Auto-Negotiation	Auto-selects best mode	1 Gbps switch connects at 100 Mbps full-duplex if needed
Auto-MDIX	Detects and adjusts cable type	Works with straight or crossover cables
Manual Configuration	Admin sets speed/duplex manually	Enterprise switches only
Cost Factor	More control = higher cost	Managed vs. unmanaged switches

6.5 Network Switches

Switches can forward frames using different methods, depending on how much of the frame they read before forwarding it. These methods affect speed, accuracy, and error handling.

1. Cut-Through Switching

- The switch reads only the first 12 bytes of the frame (just enough to get the source and destination MAC addresses).
- As soon as it knows where the frame should go, it immediately forwards the frame to the destination port - without reading the rest of the frame.
- Advantages
 - Fastest method minimal delay.
 - Ideal for networks where errors are rare and speed is critical.
- Disadvantages
 - The switch forwards all frames, even those with errors.
 - Wastes bandwidth when bad frames are forwarded and later discarded by the destination device.
- Example
 - A frame arrives at the switch. The switch reads the destination MAC and sends it out the right port instantly.
 - If the frame was corrupted, the destination still receives it but discards it.
- Speed: Fastest
- Error Handling: Forwards all errors

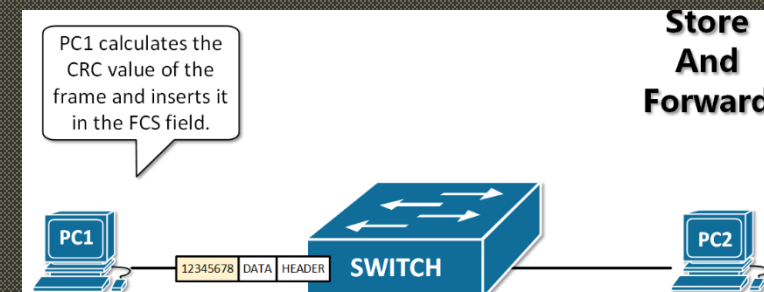


<https://www.networkacademy.io/ccna/ethernet/store-and-forward-vs-cut-through-switching>

6.5 Network Switches

2. Store-and-Forward Switching

- The switch reads the entire frame and stores it in memory.
- It checks the Frame Check Sequence (FCS) - a checksum at the end of the frame - to ensure there are no errors.
- Only error-free frames are forwarded.
- Advantages
 - No corrupted frames are sent - conserves bandwidth.
 - Provides better network reliability and stability.
 - Can handle different speed ports (e.g., 100 Mbps to 1 Gbps) easily.
- Disadvantages
 - Slower than cut-through: it must read the entire frame before forwarding.
 - Requires more memory and processing time.
- Example
 - The switch receives a full 1518-byte Ethernet frame.
 - It checks the FCS - if valid, it forwards it.
 - If invalid, it discards it.
- Speed: Slowest
- Error Handling: No error frames forwarded

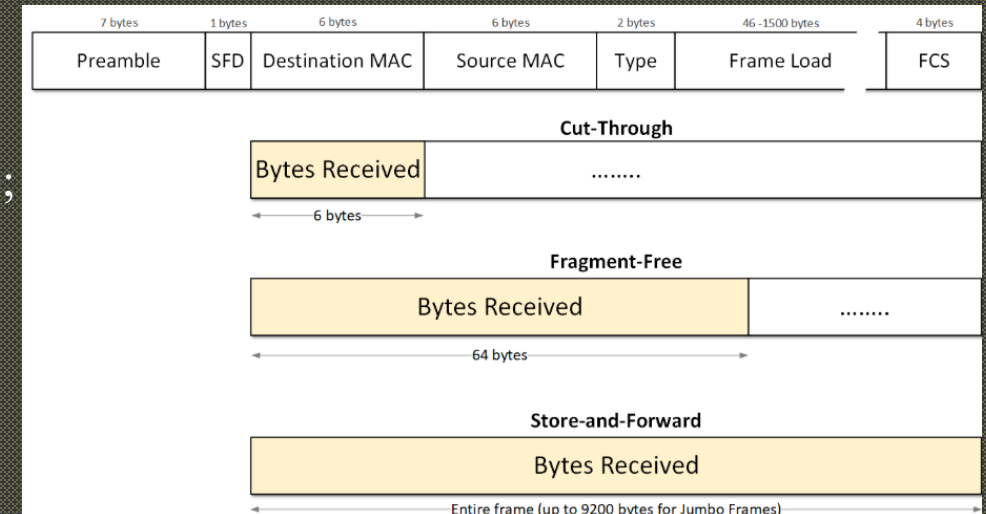


<https://www.networkacademy.io/ccna/ethernet/store-and-forward-vs-cut-through-switching>

6.5 Network Switches

3. Fragment-Free Switching

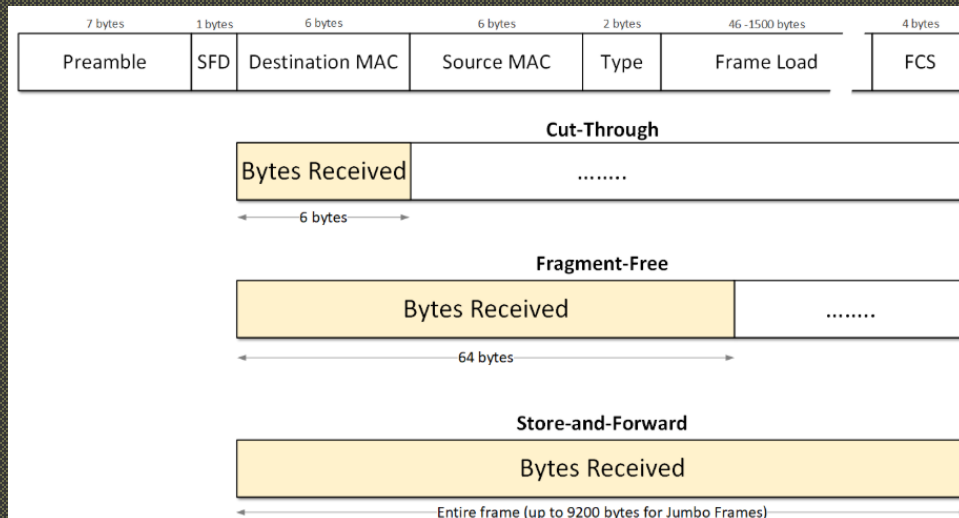
- The switch reads the first 64 bytes of the frame before forwarding.
- This ensures that the frame is not a collision fragment (damaged due to errors or collisions).
- After confirming it is at least 64 bytes, it forwards the frame.
- Advantages
 - Faster than store-and-forward, but more reliable than cut-through.
 - Filters out collision fragments (undersized frames).
- Disadvantages
 - Does not detect all types of errors - only filters undersized (fragmented) frames.
 - Not as accurate as store-and-forward.
- Example
 - Ethernet minimum frame size = 64 bytes.
 - The switch checks the first 64 bytes. If valid, it forwards the frame; if it is shorter (a fragment), it discards it.
- Speed: Medium
- Error Handling: Forwards all except undersized (fragmented) frames.



6.5 Network Switches

4. Adaptive Switching (Hybrid Approach)

- Some high-end switches can automatically switch between methods based on error rate:
 - Operate in cut-through mode for maximum speed.
 - If too many errors occur → automatically change to store-and-forward to reduce bad frames.
 - When errors decrease → switch back to cut-through for better performance.
- Example
 - A gigabit switch runs in cut-through mode.
 - It detects frequent bad frames (due to a faulty NIC).
 - The switch automatically switches to store-and-forward mode until errors reduce.



Switching Method	Performance (Speed)	Error Handling	Checks Entire Frame?	Best Used When...
Cut-through	Fastest	Forwards all errors	No	Network is reliable and low error rate
Fragment-free	Medium	Filters undersized frames only	No	Occasional collision issues
Store-and-forward	Slowest	Forwards only error-free frames	Yes	Network with frequent errors

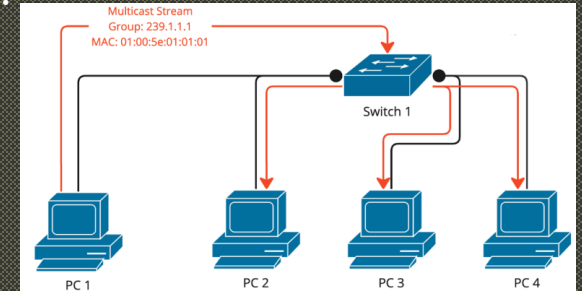
6.5 Network Switches

- High-end or managed switches (also called smart switches) do more than just forward frames. They include advanced features to make networks faster, more secure, and more reliable.

- Multicast Processing
- Spanning Tree Protocol (STP)
- Virtual Local Area Networks (VLANs)
- Port Security

1. Multicast Processing

- A multicast frame is a special type of network message sent to multiple devices, but not everyone on the network.
- There are two main ways a switch can process multicast traffic:
 - a. Treat as Broadcast (Flooding Method)
 - The switch forwards multicast frames to all ports (except the one it came from).
 - Used by low-end switches or those not configured for multicast.
 - Disadvantage: Wastes bandwidth, because all devices receive the data even if they don't need it.
 - b. Forward Only to Registered Ports (Efficient Method)
 - The switch sends multicast frames only to ports that need them - ports that have “registered” interest in that multicast address.
 - This is achieved through IGMP Snooping (Internet Group Management Protocol).
 - The switch listens to IGMP messages and learns which devices are part of which multicast group.
- Frames are only forwarded to those devices.

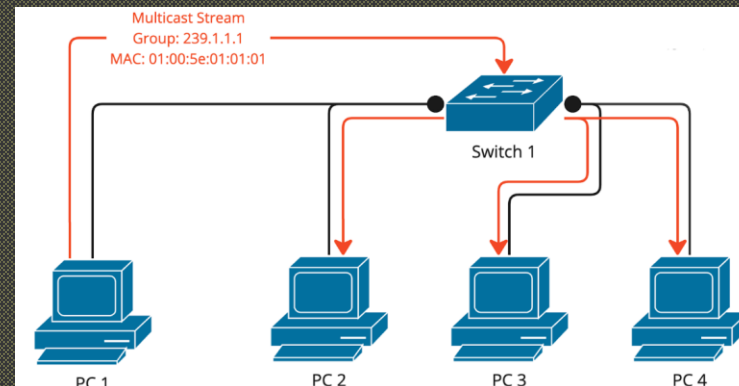


https://documentation.meraki.com/MS/Other_Topics/Layer_2_Multicast_and_Troubleshooting

6.5 Network Switches

1. Multicast Processing

- How the Switch Recognizes Multicast Addresses
 - Destination Multicast MAC addresses always begin with: 01:00:5E
 - The remaining part of the address (last 24 bits) identifies the specific multicast group/application.
 - When a switch detects a frame with a destination MAC starting with 01:00:5E, it:
 - Registers that port as part of the multicast group.
 - Forwards future multicast frames for that group only to those registered ports.
- Registering and Unregistering Multicast Memberships
- Computers (clients) use IGMP messages to tell the switch:
 - “I want to join this multicast group.” → switch registers the port.
 - “I am leaving this group.” → switch removes the port from the group.



6.5 Network Switches

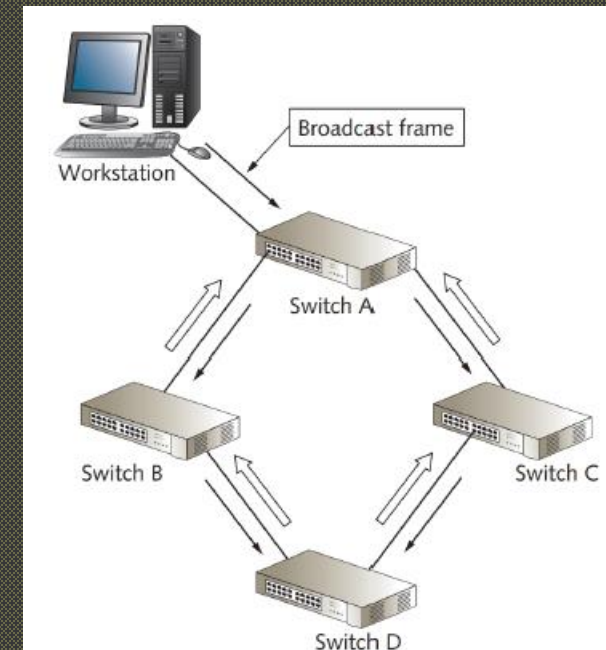
2. Spanning Tree Protocol (STP)

STP is a loop-prevention mechanism used in Ethernet networks with redundant switch connections.

- Redundancy is good - it provides backup paths in case of link or switch failure.
- But redundancy can also cause switching loops, which can crash the network.

A switching loop occurs when switches are connected in such a way that frames can circulate endlessly.

- If Switch A, B, C, and D are interconnected in a redundant loop:
 - A broadcast frame from a PC attached to Switch A is sent to B and C.
 - B and C send the broadcast to D.
 - D sends it back to B and C.
 - B and C send it back again to A...
 - And the process repeats infinitely!
- This creates what is called a broadcast storm - continuous circulation of broadcast frames that:
 - Floods all ports and bandwidth,
 - Overloads switches and CPUs,
 - Causes network slowdown or complete failure.



6.5 Network Switches

2. Spanning Tree Protocol (STP)

How STP Solves the Problem

- STP allows switches to communicate with each other to detect loops.
- It then blocks one or more ports to break the loop, but keeps them as standby paths, which means:
 - When STP detects potential loops, one port on a switch goes into blocking mode:
 - It does not forward data frames.
 - It remains active for monitoring and redundancy.
 - If a link or switch fails: STP reactivates the previously blocked port to restore connectivity - providing automatic redundancy.

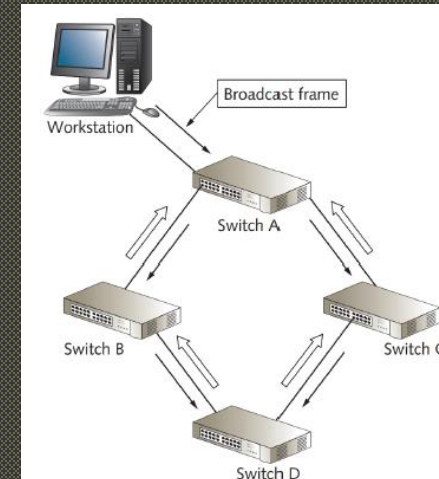
BPDUs (Bridge Protocol Data Units): Special STP messages that switches exchange to share network information and detect loops.

STP Convergence Time

- STP can take 30 seconds or more to adjust (converge) after a network change.
- During this time, some links may be temporarily unavailable.

Rapid Spanning Tree Protocol (RSTP - IEEE 802.1w)

- RSTP is a faster version of STP.
- It can reconverge in less than 10 seconds.
- RSTP is used if all connected switches support it.
- Provides the same protection against loops but with faster recovery.



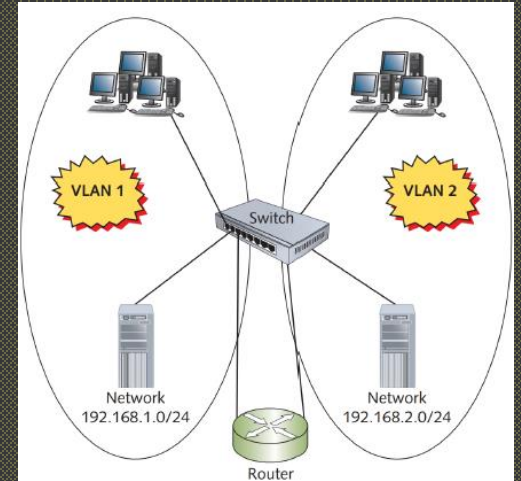
6.5 Network Switches

3. Virtual Local Area Networks (VLANs)

- A VLAN is a logical subdivision of a physical network (a switch).
- It allows you to separate one physical switch into multiple logical networks.
- Each VLAN acts like a separate broadcast domain - just as if you had multiple physical switches.

VLANs are used to:

- Improve security → Devices in different VLANs cannot see each other's broadcasts.
- Enhance performance → Reduces unnecessary broadcast traffic.
- Simplify network management → Group users or devices logically (by function, not location).
- Add flexibility → You can move devices or create virtual workgroups easily.
- Each VLAN is a separate broadcast domain.
- Devices in the same VLAN can communicate directly (Layer 2).
- Devices in different VLANs need a Layer 3 device (a router or Layer 3 switch) to communicate.



Example:

There is one 48-port switch divided it into:

- VLAN 10: Administration
- VLAN 20: Students
- VLAN 30: Faculty
- Now, devices in different VLANs cannot communicate directly without a router.

6.5 Network Switches

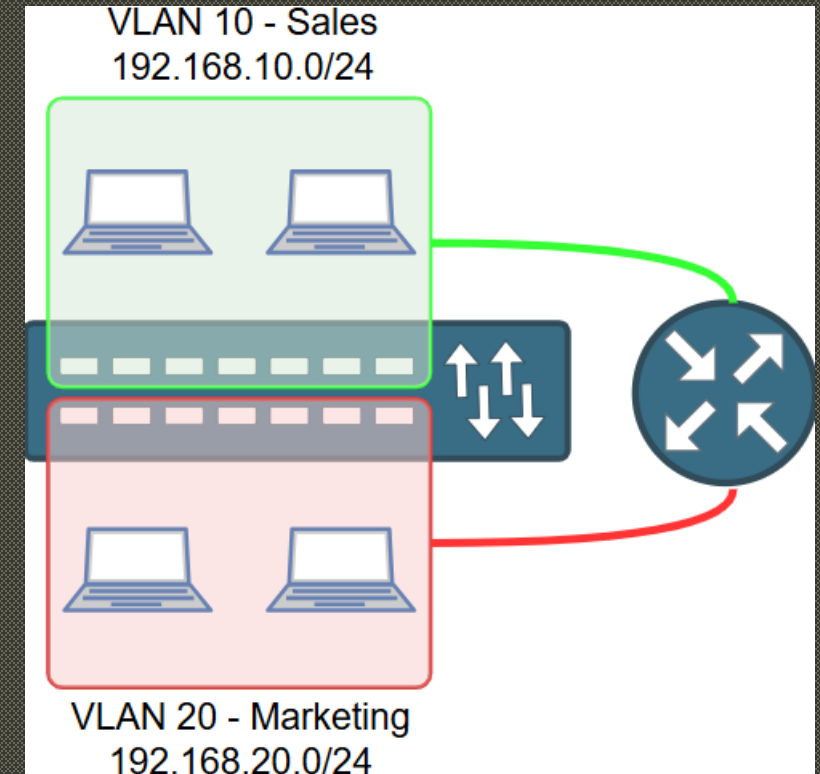
3. Virtual Local Area Networks (VLANs)

Communication Between VLANs

- VLANs cannot communicate directly.
- A router (or Layer 3 switch) is required to route packets between VLANs.
- This process is called Inter-VLAN Routing.
- Example:
 - PC1 in VLAN 10 wants to talk to PC2 in VLAN 20.
 - The switch sends traffic to the router, which routes it back to VLAN 20.

VLANs and IP Addressing

- Devices in different VLANs must have different IP networks.
 - VLAN 10 → 192.168.10.0/24
 - VLAN 20 → 192.168.20.0/24
- If a device is moved to another VLAN, its IP address must change to match that VLAN's network.
- Example:
 - If a PC from VLAN 10 (192.168.10.x) moves to VLAN 20,
 - It must now use 192.168.20.x to communicate.



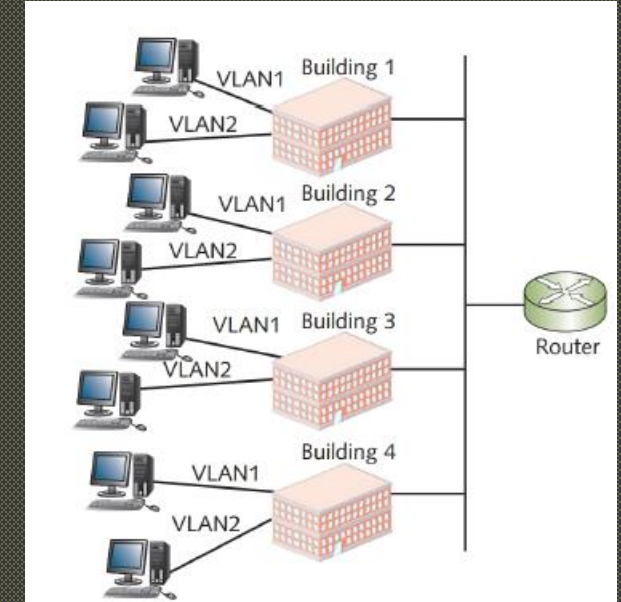
<https://codingpackets.com/blog/network-notes-vlan/>

6.5 Network Switches

3. Virtual Local Area Networks (VLANs) VLAN Trunks

Why We Need VLAN Trunks

- If each VLAN were connected separately between switches, you would need one physical cable per VLAN - for example:
 - VLAN 10 (Admin) → one cable between switches
 - VLAN 20 (Engineering) → another cable
 - VLAN 30 (Operations) → another cable
- If you had 10 VLANs, that would mean 10 cables and 20 ports just to interconnect two switches - totally impractical.
 - VLAN trunks allow a single cable (and port) to carry traffic for multiple VLANs by using VLAN tags.



What Is a VLAN Trunk Port

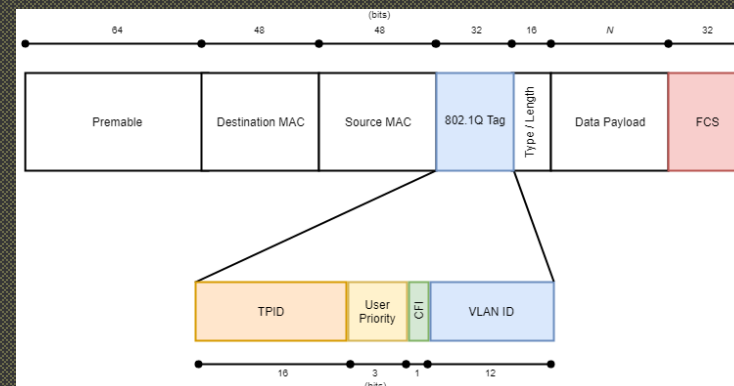
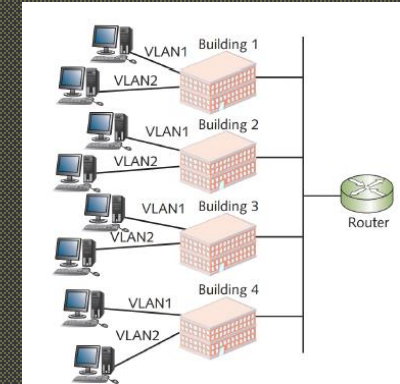
- A VLAN trunk port is a switch port configured to carry traffic from multiple VLANs. It does not belong to just one VLAN; instead, it uses a VLAN tag in each frame to tell which VLAN the frame belongs to.
- Both ends of the link (e.g., Switch-Switch or Switch-Router) must be configured as trunk ports and use the same trunking protocol - typically IEEE 802.1Q (the industry standard).

6.5 Network Switches

3. Virtual Local Area Networks (VLANs) VLAN Trunks

How IEEE 802.1Q Tagging Works

- When a frame leaves a VLAN trunk port:
 - The switch inserts a small tag (4 bytes) into the Ethernet frame header.
 - This tag contains the VLAN ID (e.g., 10, 20, 30).
 - The receiving switch reads the tag, identifies which VLAN it belongs to, and forwards it accordingly.
 - Before the frame is sent to a device on an access port, the switch removes the tag so end devices never see it.
- Think of it like this:
 - The VLAN tag acts as a shipping label on the frame while it travels between switches - it tells the next switch,
 - “This frame belongs to VLAN 20!”
 - When the frame reaches its destination network (the correct VLAN), the tag is peeled off before delivery.



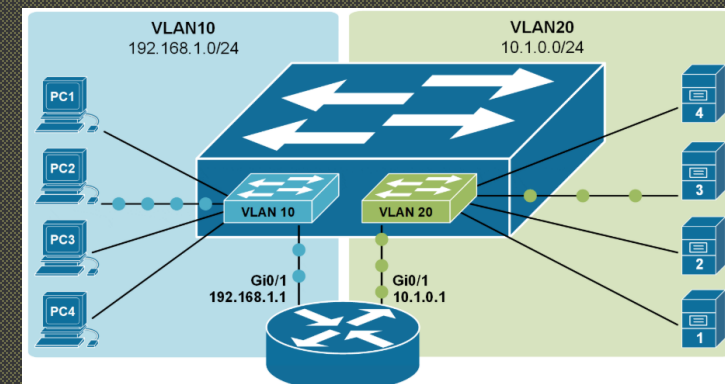
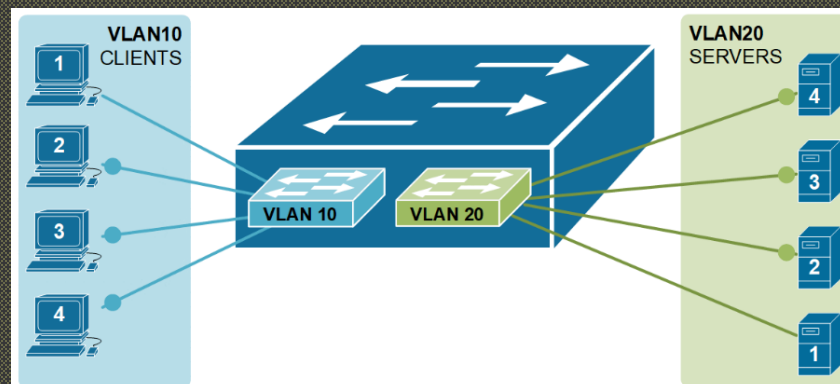
Field	Purpose
TPID	Contains the HEX value 0x8100 which signals the receiver the presence of an 802.1Q tag in the frame header.
User Priority	Defined in the 802.1p standard to align with CoS marking used in upper layers such as IP Precedence and MPLS EXP.
Canonical Format Indicator	Used to associate the VLAN ID field with either ethernet (0) or token ring (1). Should always be set to 0 for ethernet.
VLAN ID	A VLAN ID between 1 and 4094.

6.5 Network Switches

3. Virtual Local Area Networks (VLANs)

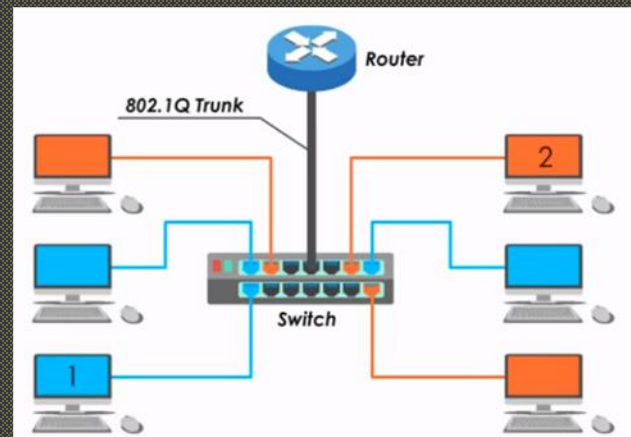
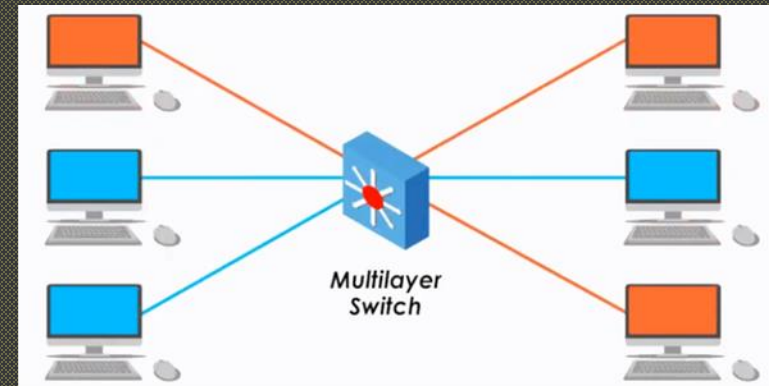
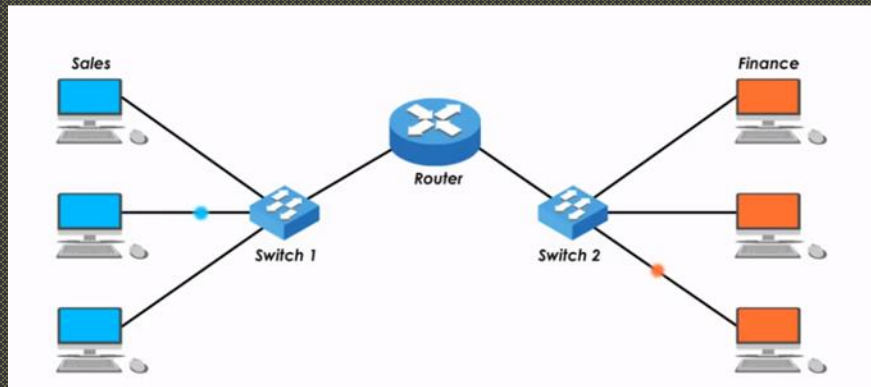
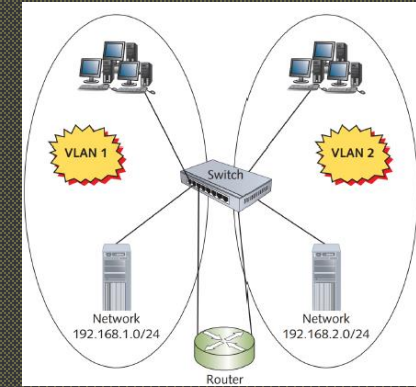
VLANs are powerful tools for: Network organization, Security segmentation, Broadcast control. But they should be used strategically, not excessively. Always plan VLAN design based on real traffic flow, resources, and hardware capability; otherwise, you can end up with higher costs, slower performance, and unnecessary complexity.

Factor	Description	Impact / Example
Network Complexity	More VLANs → more logical networks	Harder to manage and troubleshoot
Router Requirement	Needed for inter-VLAN communication	Extra cost and setup
Cost	Requires managed switches or routers	Higher network cost
Subnetting	Each VLAN needs a unique IP range	More planning needed
Reconfiguration	Devices may need new IPs	Time-consuming
Performance	Inter-VLAN routing adds delay	Slower communication if poorly designed
Planning	VLANs should match usage and traffic	Prevents inefficiency



6.5 Network Switches

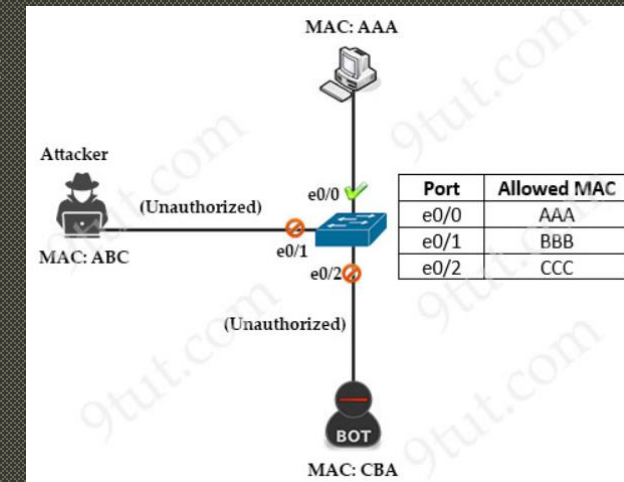
3. Virtual Local Area Networks (VLANs)



6.5 Network Switches

4. Switch Port Security

- Port security is a network access control feature available on managed switches.
- It helps protect the Local Area Network (LAN) by restricting which devices can connect to a switch port.
- This feature is particularly important in public or semi-public environments.
- Without port security, anyone could plug in a laptop or other device into a network jack and:
 - Infect the network with viruses or malware
 - Run hacker tools to capture traffic
 - Access sensitive internal resources
- Port security enables network administrators to:
 - Limit the number of MAC addresses that can connect to a specific switch port.
 - Specify which MAC addresses are allowed to use that port.
 - Take automatic actions if an unauthorized device tries to connect.
- When port security is enabled on a switch:
 - The switch learns the MAC address of the device connected to the port.
 - That MAC address is recorded in the switch's secure address table.
 - If another device (with a different MAC address) tries to use that port:
 - The switch detects the violation, and
 - Takes an action based on its configuration.



<https://www.9tut.com/port-security-tutorial>

Action Type	Description
Protect	Drops packets from unknown MAC addresses but keeps the port active.
Restrict	Drops packets and sends a notification (SNMP trap or log message).
Shutdown	Disables the port completely until manually re-enabled by an administrator.

6.5 Network Switches

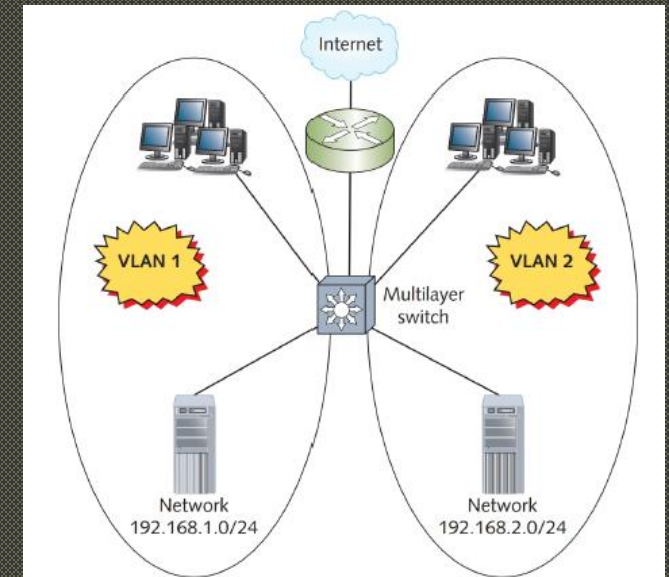
- A multilayer switch is a network switch that can also perform routing (Layer 3) functions.
- It combines the speed of a switch with the intelligence of a router.
- Operates at both:
 - Layer 2 (Data Link) – Switching frames within the same VLAN.
 - Layer 3 (Network) – Routing packets between different VLANs or subnets.
- Used inside large networks (LANs or campus networks) to:
 - Route traffic between VLANs (Inter-VLAN Routing).
 - Reduce network latency by keeping routing inside the switch.
- Not typically used for Internet/WAN connections that's still done by a traditional router.

Without a Multilayer Switch:

- VLAN 10 and VLAN 20 cannot communicate directly.
- You need a router connected to both VLANs to route packets.

With a Multilayer Switch:

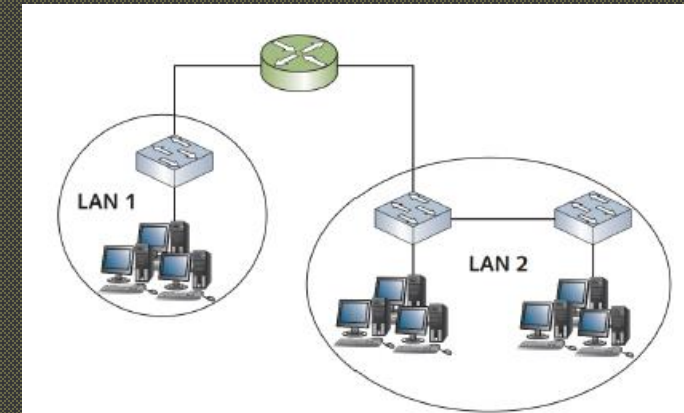
- The switch itself has virtual interfaces for each VLAN (called SVIs – Switched Virtual Interfaces).
- Example:
 - VLAN 10 SVI → 192.168.10.1
 - VLAN 20 SVI → 192.168.20.1
- When a PC in VLAN 10 sends data to VLAN 20:
 - The multilayer switch receives the packet.
 - It checks the VLAN tag and destination IP.
 - It routes the packet internally (Layer 3 process).
 - The packet stays inside the switch — no need to go to an external router.



6.6 Routers

What is a Router?

- A router is a network device that connects multiple LANs (Local Area Networks) together.
- While hubs and switches connect computers inside one LAN, a router connects entire LANs to form a bigger network, called an internetwork.
- The Internet itself is made up of many LANs connected by routers.
- Example:
 - LAN 1: Office computers on the 1st floor
 - LAN 2: Office computers on the 2nd floor
- A router connects these two LANs so data can flow between them.



Router Connections

- Routers usually have two or more network ports:
 - Each port connects to a different LAN or network device (like a switch or hub).
 - Example:
 - Port 1 connects to the Office LAN 1 (via a switch)
 - Port 2 connects to the Office LAN 2 (via a switch)
 - Think of a router as a “bridge” between separate networks.
- Hubs and Switches: Connect devices within one LAN.
 - Routers: Connect different LANs together (LAN-to-LAN or LAN-to-Internet).
 - Routers use IP addresses and routing tables to make intelligent forwarding decisions.
 - Switches use MAC addresses and switching tables to forward frames locally.
 - Routers stop broadcast traffic from spreading beyond the LAN, improving performance.

6.6 Routers

Router Function - How It Works

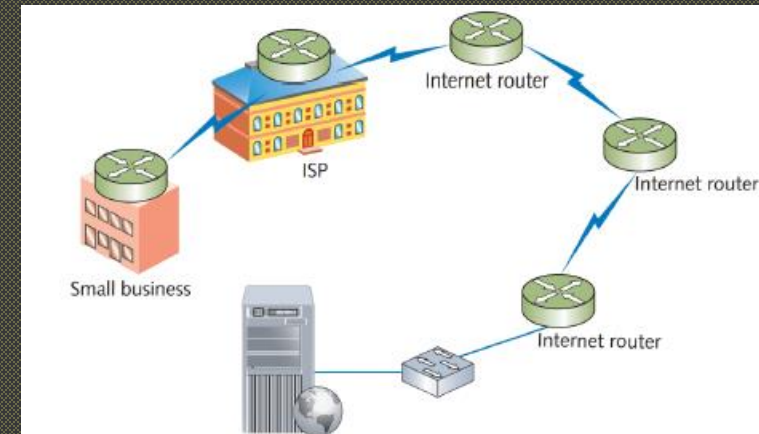
- Routers receive data packets from one network and decide where to send them next.
- They use routing tables to determine the best path for the data.
- If data must cross several routers to reach its destination (like on the Internet), each router forwards the packet to the next one until it arrives.
- *Example:*
 - Switch = local traffic director (controls traffic within one building).
 - Router = city traffic controller (directs traffic between different cities).
- You want to get into Humber.ca from your laptop (LAN of Small business) → Router sends it to your ISP → Then through multiple routers across the Internet → Finally, to your Humber's LAN (LAN of Humber server).

Routers Create Internetworks

- Each LAN is a small “island” of computers.
- Routers connect these islands together to form a continent of networks - the Internet.
- This interconnection between many LANs via routers is called an internetwork.

Routing Table

- A routing table is like a map the router uses to decide where to send data packets.
- It lists known networks and the best path (next router or interface) to reach each one.
- The router constantly updates this table as it learns new routes.
- Example: If Router A does not know how to reach Network X, it forwards the packet to Router B, which does.



6.6 Routers

Problem: Broadcast Traffic

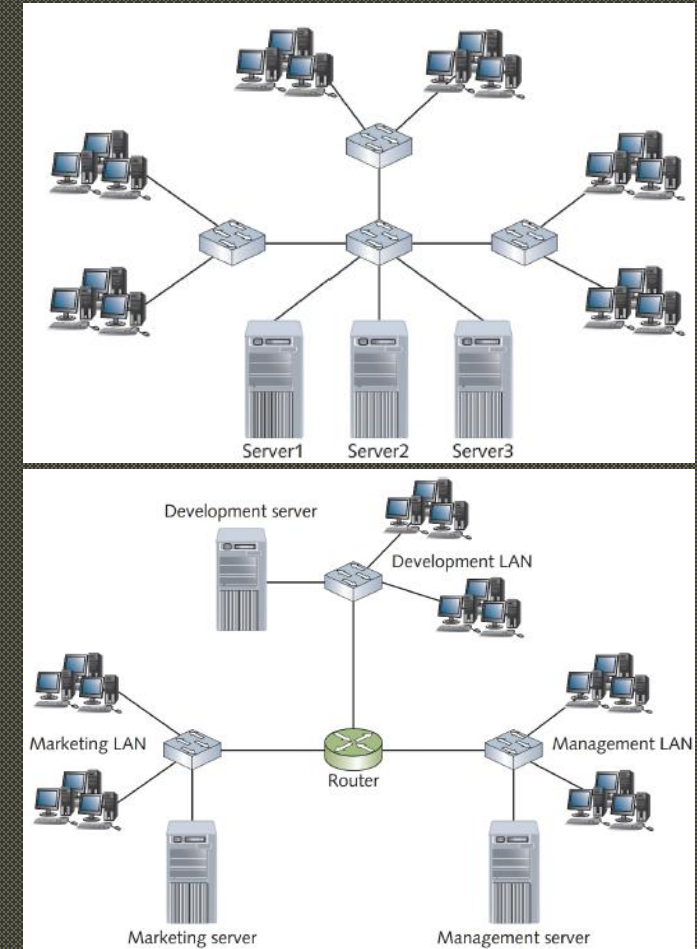
- Every time a computer sends a broadcast, the switch must forward it to all computers.
- Each computer's CPU must stop what it's doing to process that broadcast.
- When there are many broadcasts, the network becomes less efficient - communication slows down.

Solution: Divide the LAN into Smaller LANs

- To reduce congestion, divide the large LAN into smaller LANs (subnetworks).
- Each smaller LAN has computers that share similar needs (e.g., by department).
- Each LAN has its own switch and server for local communication.

Role of the Router

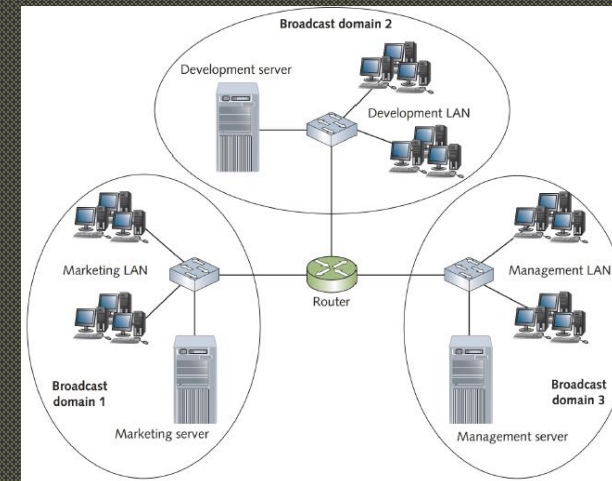
- A router connects these smaller LANs together.
- The router acts like a messenger:
 - It forwards messages meant for another LAN.
 - It blocks broadcasts, so announcements stay within their LAN.
- This makes communication more efficient - fewer unnecessary messages.



6.6 Routers

- A broadcast domain is the area of a network where broadcast messages are shared.
- When one computer sends a broadcast frame, all devices in the same broadcast domain receive it.
- Routers do not forward broadcast frames.
- This means each router interface acts as a boundary for a broadcast domain.
- So, every router port connected to a different LAN creates a separate broadcast domain.
- If a network has no routers, all devices belong to one big broadcast domain - every broadcast reaches everyone.
- If a network has one router with two interfaces, there are two broadcast domains - one on each side of the router.
- Smaller broadcast domains = less network congestion.
- Routers limit broadcast traffic to only the devices that need it.
- This helps networks run faster and more efficiently.

Function	Switch	Router
Works with	MAC addresses (physical)	IP addresses (logical)
Data unit	Frame	Packet
Table type	Switching table	Routing table
Connects	Devices inside a LAN	Different LANs together

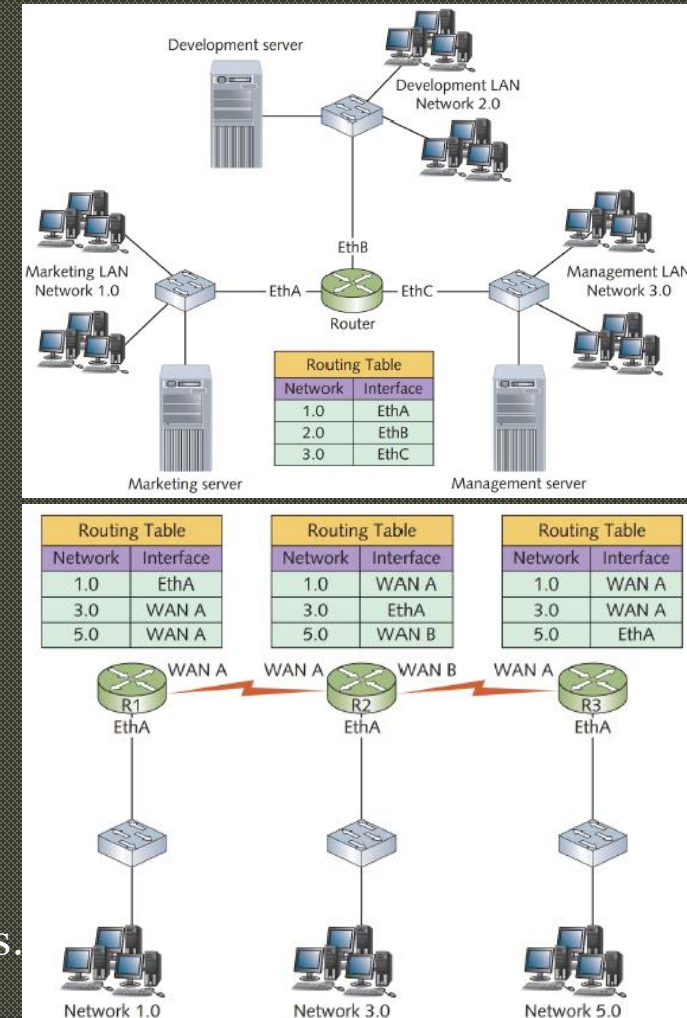


6.6 Routers

- Routers are smart devices that decide where data packets go between networks.
- They do this by using IP addresses and routing tables.
- A router has two or more interfaces (ports).
 - Each interface connects to a different network.
- When a router receives a packet, it checks the destination IP address.
- The router looks in its routing table to find which interface will best reach the destination.
- It then forwards the packet out through that interface.

Example: Imagine a router with three interfaces:

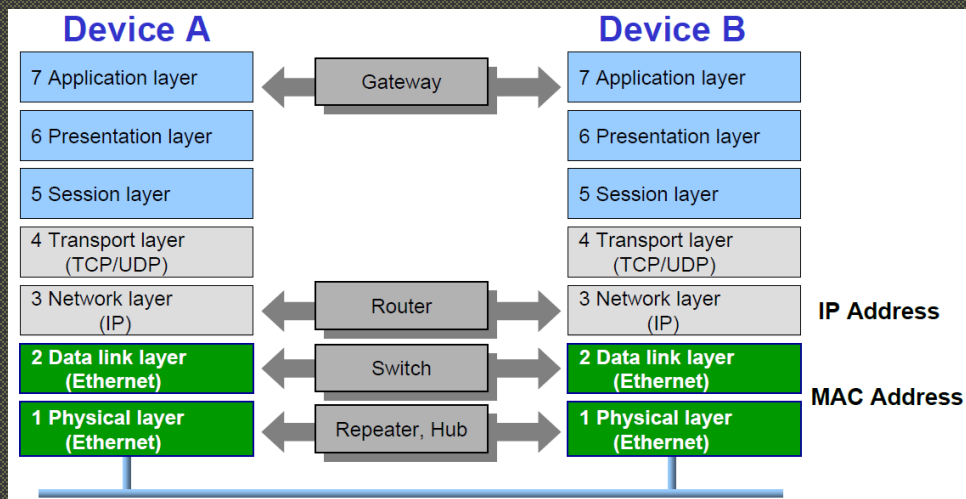
- EthA → Network 1.0; EthB → Network 2.0; EthC → Network 3.0
- If a computer in Network 1.0 sends data to Network 3.0:
- The router checks its routing table, sees Network 3.0 → EthC,
- Forwards the packet out of EthC to reach its destination.
- If the router does not have a direct connection to the destination network:
- It forwards the packet to another router that is closer to the destination.
- Each router repeats this process until the packet reaches the target network.
- This process is called routing, the step-by-step delivery of packets through multiple routers.



6.6 Routers

Default Gateway

- Router: A device that determine the optimal path along which network traffic should be forwarded. Routers allow different networks to communicate with each other.
- Often in industry is used the term Default Gateway for the next router.
- The default gateway is the uplink router (the router connected to that LAN).



Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . : hitronhub.home
Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
Physical Address. . . . . : 94-65-9C-85-F5-93
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.0.27(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, November 3, 2020 12:24:55 PM
Lease Expires . . . . . : Saturday, November 14, 2020 9:49:47 AM
Default Gateway . . . . . : 192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DNS Servers . . . . . : 192.168.0.1
NetBIOS over Tcpip. . . . . : Enabled
```

6.6 Routers

Default Gateway

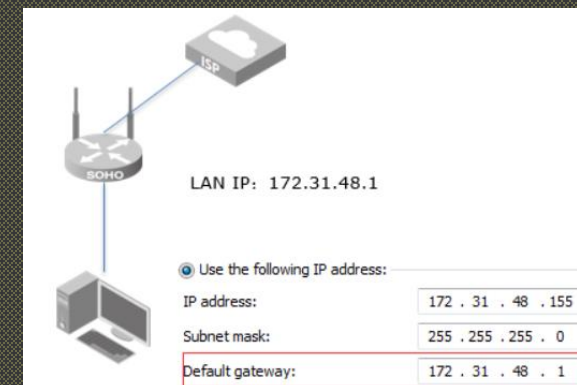
- A default gateway is the router that a computer sends data to when the destination is outside its local network (LAN).
- It tells the workstation: *If the destination is not on your LAN, send it to this router.*
- The workstation checks its IP address against the destination IP address.
- If the destination is on the same LAN:
 - The workstation gets the MAC address of the destination device.
 - Sends the frame directly over the LAN.
- If the destination is on a different network:
 - The workstation gets the MAC address of the router (default gateway).
 - Sends the frame to the router, which then forwards it to the correct network.
- If Default Gateway is not set:
 - Computer can only communicate with devices on the same LAN.
 - Packets destined for other networks cannot leave the LAN.

Example 1 – Same Network

- Workstation IP: 192.168.1.10
- Destination IP: 192.168.1.20
- Both are on 192.168.1.0/24 network → Workstation sends directly to the destination MAC.

Example 2 – Different Network

- Workstation IP: 192.168.1.10
- Destination IP: 192.168.2.15
- Different network → Workstation sends the frame to the default gateway (router).
- Router forwards the packet to Network 192.168.2.0 and eventually to the destination computer.



6.6 Routers

- Routers operate at Layer 3 (Network Layer) of the OSI model.
- They work with packets as their main data unit (PDU).
- Routers connect different logical networks to form one larger internetwork (a network of networks).

Example: At Humber College, one router can connect:

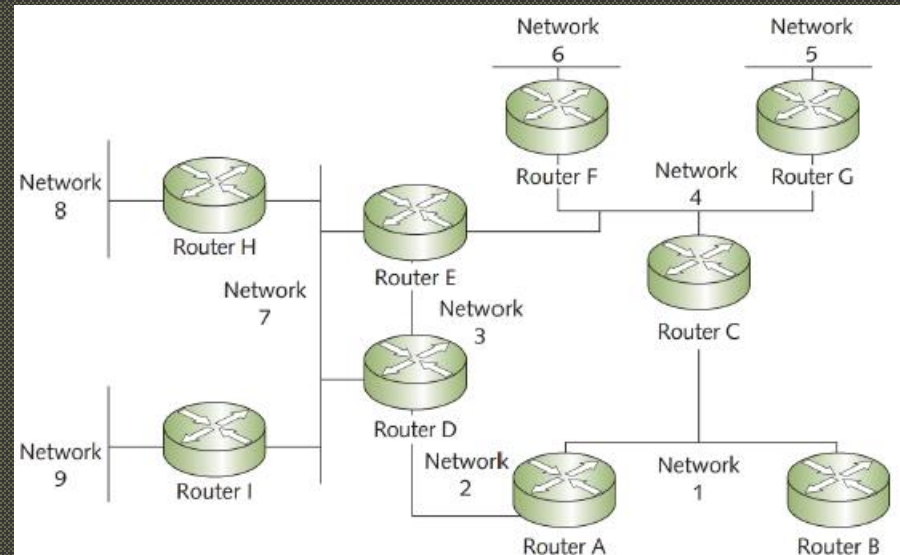
- The Campus LAN,
- The Wi-Fi Network, and
- The Internet - allowing them to communicate properly.

- Routers forward packets between networks, not within the same network.
- They block broadcast frames - broadcasts stay inside their own LAN.
- This prevents unnecessary traffic between networks.

Example:

If a PC in the Engineering VLAN sends a broadcast frame, the router doesn't forward it to the Business VLAN or to the Internet.

This avoids a "broadcast storm."



6.6 Routers

- The Internet is made up of thousands of interconnected networks.
- Routers connect all these networks without forwarding broadcast frames.
- Each network connected to a router is its own broadcast domain.
- This means broadcast traffic stays local, improving efficiency and performance.

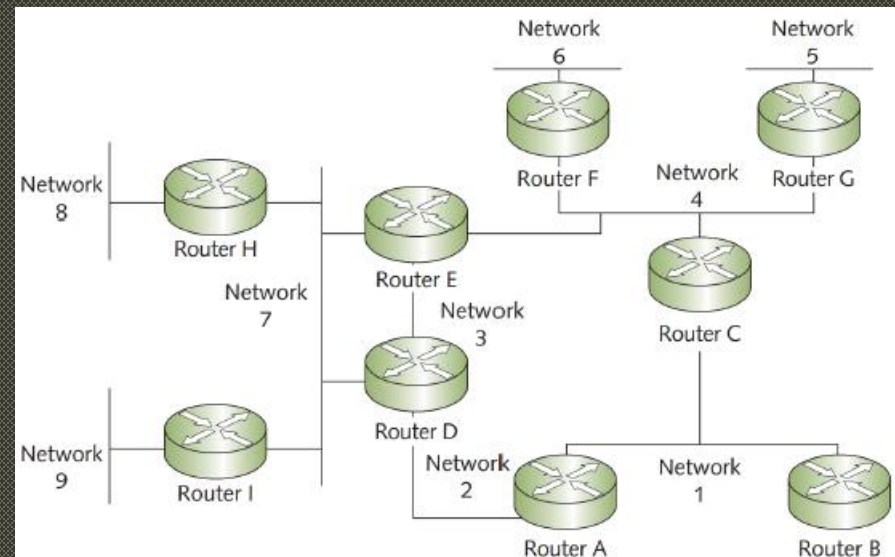
- Routers Support Multiple Paths
- Routers can use multiple network paths between two locations.

These paths provide:

- Fault tolerance → If one link fails, another is used.
 - Load sharing → Traffic can be distributed across several links to reduce congestion.
- Routers can filter or block packets based on rules set by the network administrator.
 - This is done using Access Control Lists (ACLs).
 - Helps improve security and traffic management.

For example, a router might:

- Allow traffic from Humber's internal network to the Internet.
- Block incoming traffic from unknown external sources.



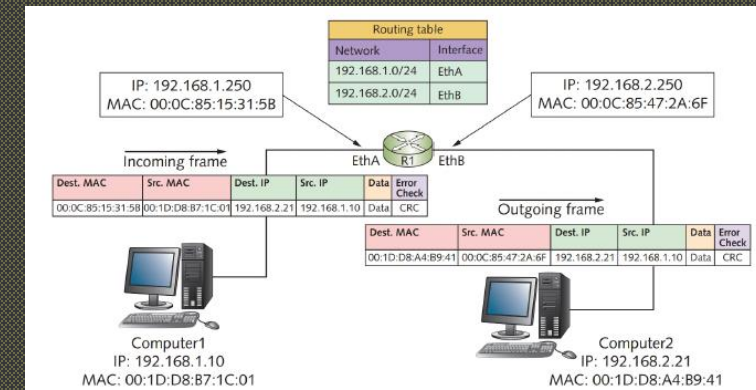
6.6 Routers

- A router needs two or more interfaces (ports) to connect different networks.
- Each interface belongs to a different network.
- Each interface has:

- A MAC address (Layer 2 – physical address)
- An IP address (Layer 3 – logical address)

Example: A simple router connecting two networks:

- Interface R1-Gi0/1 → 192.168.1.254 (LAN left side)
- Interface R2-Gi0/1 → 192.168.2.254 (LAN right side)



- Each router interface acts like a network card (NIC) on a computer.
- It receives frames, processes them, and sends them to another network.
- The IP protocol is bound to each interface, allowing Layer 3 communication.

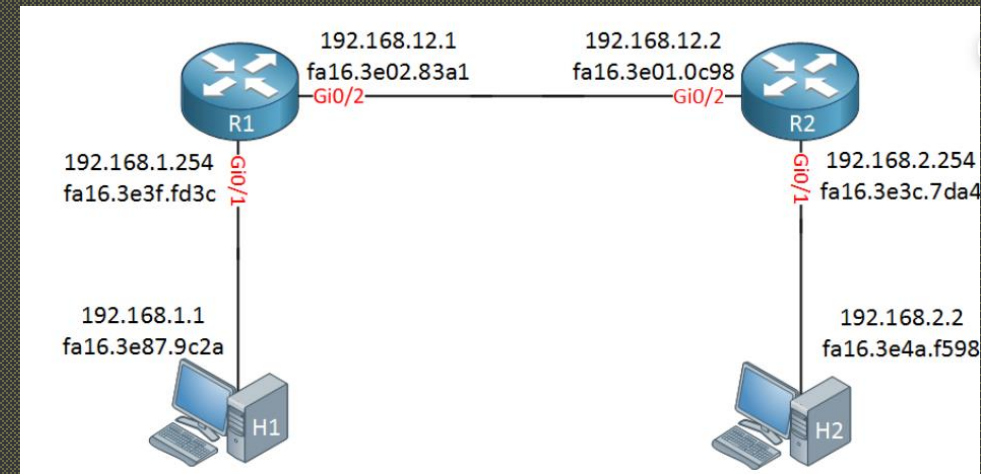
Example:

A router with two interfaces acts like two computers, each connected to a different network segment.

- A frame arrives at one of the router's interfaces.
- The router checks the destination MAC address:
 - If it matches the router's interface MAC → the router processes the frame.
 - If it does not match → the frame is discarded (it is not meant for this router).

Example:

A frame sent to MAC fa:16:3e:3f:fd:3c will only be read by the router interface with that MAC.



6.6 Routers

- Once the router confirms the frame is for it, it removes the Layer 2 header and trailer.
- What is left is the Layer 3 packet (IP packet).
- The router now looks at the destination IP address.
- The router examines the destination IP address in the packet.
 - If the destination IP matches the router's own IP → it is for the router itself (e.g., a ping).
 - If the network ID is different → the router knows the packet must be forwarded to another network.

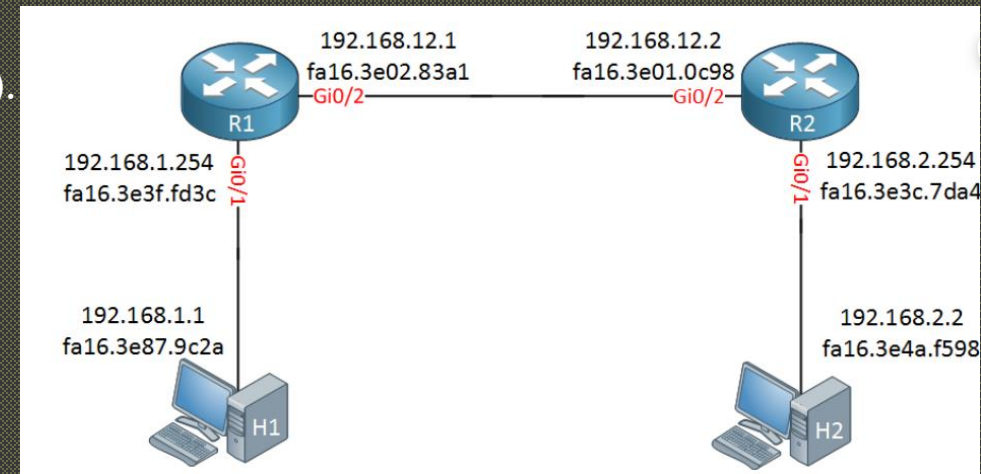
Example:

- If the router's interface is 192.168.1.254 and the packet's destination is 192.168.2.2, → different network IDs → must be routed.

- The router looks up the routing table to find the best path to the destination network.
- The routing table tells it:
 - Which interface to send the packet out of.
 - The next-hop address (next router or destination device).

For example, routing table entry:

Destination: 192.168.2.2/24 → Outgoing Interface: Gi0/2 → Next Hop: 192.168.12.2



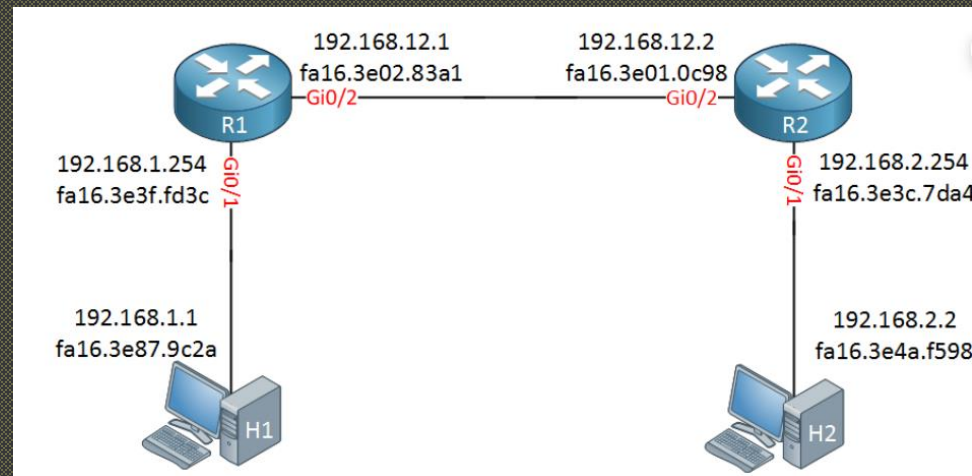
6.6 Routers

Encapsulating the Packet for the Next Hop

- The router wraps (encapsulates) the packet in a new Layer 2 frame.
- The new frame includes:
 - Source MAC: The router's outgoing interface MAC.
 - Destination MAC: The next router's or device's MAC.
- The router also creates a new CRC (Cyclic Redundancy Check) in the trailer for error checking.
- The router sends the newly encapsulated frame out through the outgoing interface.
- The packet now continues its journey - possibly through more routers - until it reaches the final destination.

Example:

- Frame on outgoing interface might look like:
- Source MAC: fa:16:3e:02:83:a1
- Destination MAC: fa:16:3e:01:0c:98



6.6 Routers

Step 1: Packet Creation on host H1 (192.168.1.1)

Subnet Check: H1's IP configuration includes its subnet mask (255.255.255.0 or /24). It performs a bitwise AND on its own IP (192.168.1.1) and the destination (192.168.2.2):

H1's network: 192.168.1.0/24.

Destination network: 192.168.2.0/24 (different from H1's).

Conclusion: Not on the local subnet, so H1 cannot send directly - must route via its default gateway (configured as 192.168.1.254, which is R1's LAN interface).

IP Packet Formation (Layer 3):

Source IP: 192.168.1.1.

Destination IP: 192.168.2.2.

Protocol: ICMP (for ping), with payload (e.g., "echo request").

TTL (Time to Live): Starts at 255 (decrements per hop to prevent loops).

ARP for Gateway MAC (Layer 2 Resolution):

H1 needs router R1's MAC address for the Ethernet frame but does not know it yet.

H1 broadcasts an ARP Request on its LAN: "Who has IP 192.168.1.254? Tell 192.168.1.1."

R1 (on the same subnet) responds with a unicast ARP Reply: "192.168.1.254 is at MAC fa16.3e3f.fd3c" (R1's Gi0/1 MAC).

H1 caches this in its ARP table (temporary, ~minutes) for future use.

Ethernet Frame Encapsulation (Layer 2):

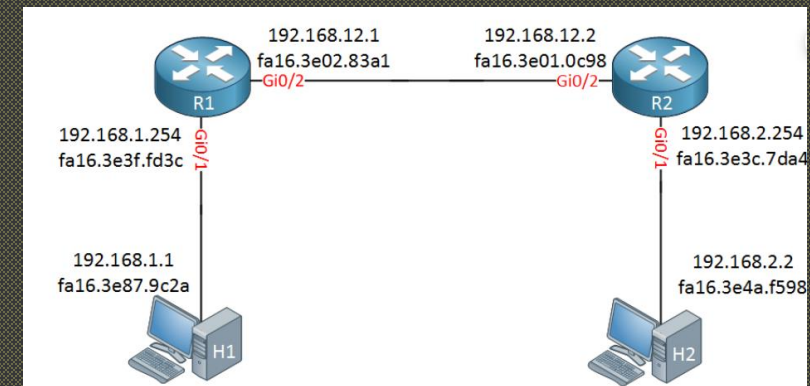
Source MAC: H1's fa16.3e87.9c2a.

Destination MAC: R1's fa16.3e3f.fd3c.

Payload: The IP packet.

Frame sent out H1's NIC over the Ethernet cable to R1's Gi0/1.

Outcome: Frame arrives at R1. No routing table lookup on H1 – it is just forwarding to the gateway.



6.6 Routers

Step 2: R1 Receives and Routes the Packet

Frame Reception on R1's Gi0/1:

R1 strips the Ethernet header, revealing the IP packet (source 192.168.1.1, destination 192.168.2.2).

Checks: TTL > 0 (yes), checksum valid, not destined for R1 itself.

Routing Table Lookup (Layer 3):

R1 consults its routing table (viewable via `show IP route`). Assuming static routes:

Directly connected: 192.168.1.0/24 via Gi0/1 (local LAN).

Directly connected: 192.168.12.0/30 via Gi0/2 (WAN link).

Static route: 192.168.2.0/24 via next-hop IP 192.168.12.2 (R2's WAN IP).

Longest prefix match: Destination 192.168.2.2 matches the 192.168.2.0/24 route.

Door Interface: Gi0/2 (WAN), next hop: 192.168.12.2.

TTL decrements to 254.

ARP for Next-Hop MAC:

On the WAN link (a point-to-point subnet), R1 ARPs for 192.168.12.2: Broadcast "Who has 192.168.12.2?"

R2 replies: "192.168.12.2 is at MAC fa16.3e01.0c98" (R2's Gi0/2 MAC).

R1 caches this.

New Ethernet Frame Encapsulation:

Source MAC: R1's WAN MAC fa16.3e02.83a1.

Destination MAC: R2's WAN MAC fa16.3e01.0c98.

Payload: Same IP packet (unchanged).

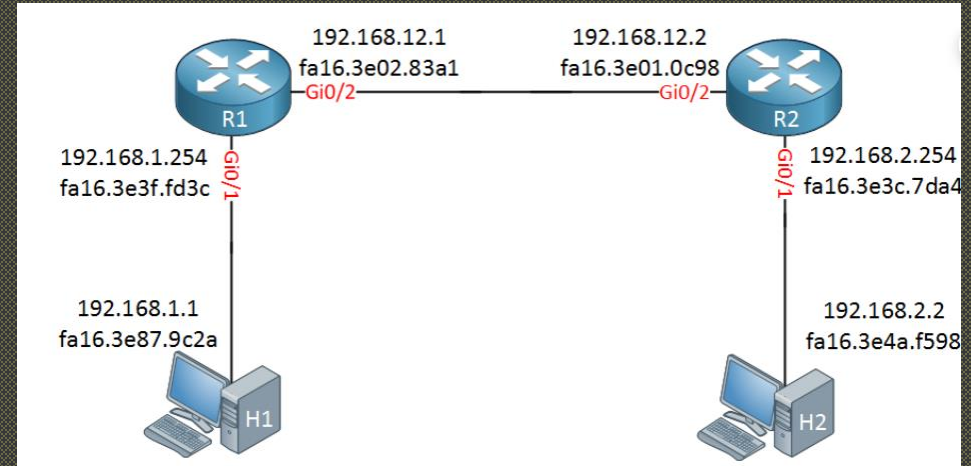
Frame sent out Gi0/2 over the WAN link (Ethernet or serial, but diagram shows Ethernet-style).

Notes:

MAC addresses change - original H1-R1 MACs are discarded; new ones are for the R1-R2 link only.

No NAT or ACLs assumed, so packet passes unmodified.

If no route existed, R1 would drop the packet and send an ICMP "Destination Unreachable" back to H1.



<https://networklessons.com/ip-routing/ip-routing-explained>

6.6 Routers

Step 3: R2 Receives and Routes the Packet

Frame Reception on R2's Gi0/2:

R2 strips the Ethernet header, extracts IP packet (source still 192.168.1.1, destination 192.168.2.2).

Checks: Valid, TTL 254 > 0.

Routing Table Lookup:

R2's routing table:

Directly connected: 192.168.12.0/30 via Gi0/2 (WAN).

Directly connected: 192.168.2.0/24 via Gi0/1 (local LAN).

Static route: 192.168.1.0/24 via next-hop 192.168.12.1 (for replies).

Match: 192.168.2.2 is local (directly connected via Gi0/1).

Door Interface: Gi0/1, no next-hop IP needed (deliver directly on LAN).

TTL decrements to 253.

ARP for Destination MAC:

R2 ARPs on its LAN: "Who has 192.168.2.2?"

H2 replies: "192.168.2.2 is at MAC fa16.3e4a.f598."

R2 caches this.

New Ethernet Frame Encapsulation:

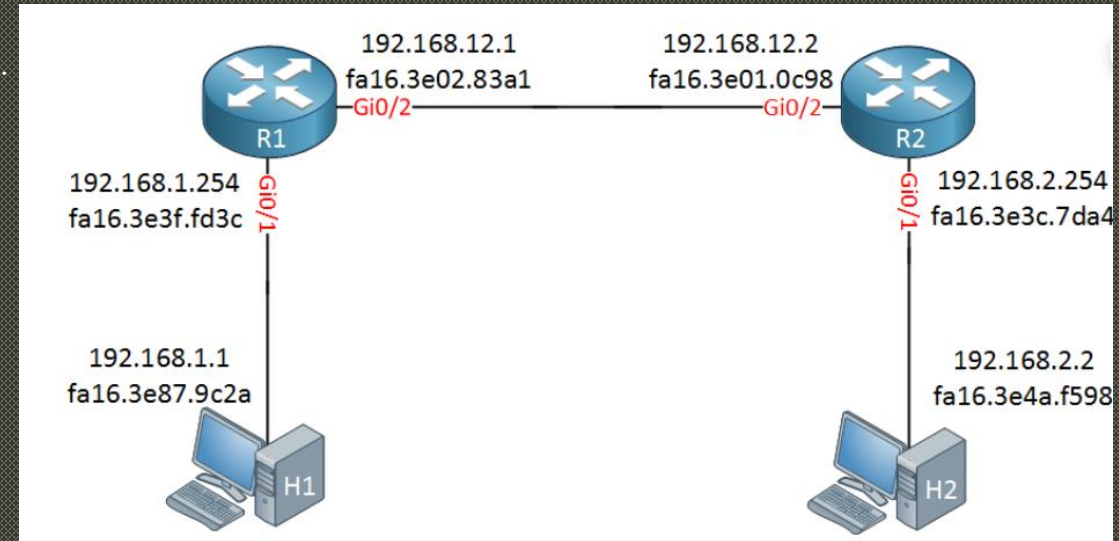
Source MAC: R2's LAN MAC fa16.3ecc.7da4 (note: diagram shows "fa16.3ec.7da4", likely abbreviated).

Destination MAC: H2's fa16.3e4a.f598.

Payload: IP packet.

Frame sent out Gi0/1 to H2 over Ethernet.

Outcome: Packet reaches H2. Total hops: 3 (H1→R1, R1→R2, R2→H2).



<https://networklessons.com/ip-routing/ip-routing-explained>

6.6 Routers

Step 4: H2 Processes the Packet and Replies

Reception on H2:

H2 strips frame, gets IP packet, processes it (e.g., replies to ping with ICMP Echo Reply).

Reply Packet: Source IP now 192.168.2.2, Dest IP 192.168.1.1.

Reverse Routing:

H2 checks: Dest 192.168.1.1 not local → Send to default gateway 192.168.2.254 (R2).

ARP for R2's MAC (fa16.3ecc.7da4).

Frame to R2's Gi0/1.

R2 Routes Reply:

Matches route for 192.168.1.0/24 → Next hop 192.168.12.1 via Gi0/2.

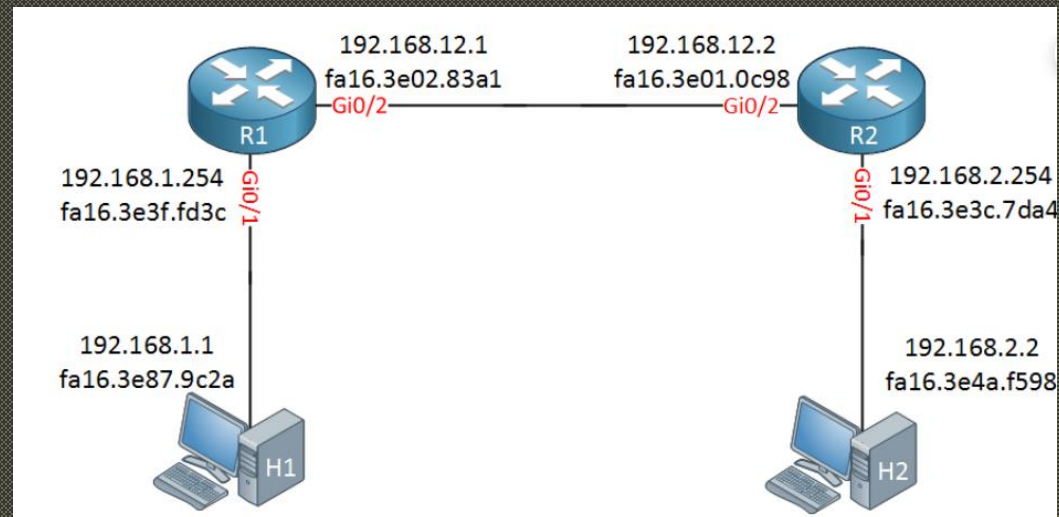
ARP for R1's WAN MAC, new frame with R2 WAN MAC → R1 WAN MAC.

R1 Routes Reply:

Matches 192.168.1.0/24 → Local via Gi0/1.

ARP for H1's MAC, new frame with R1 LAN MAC → H1 MAC.

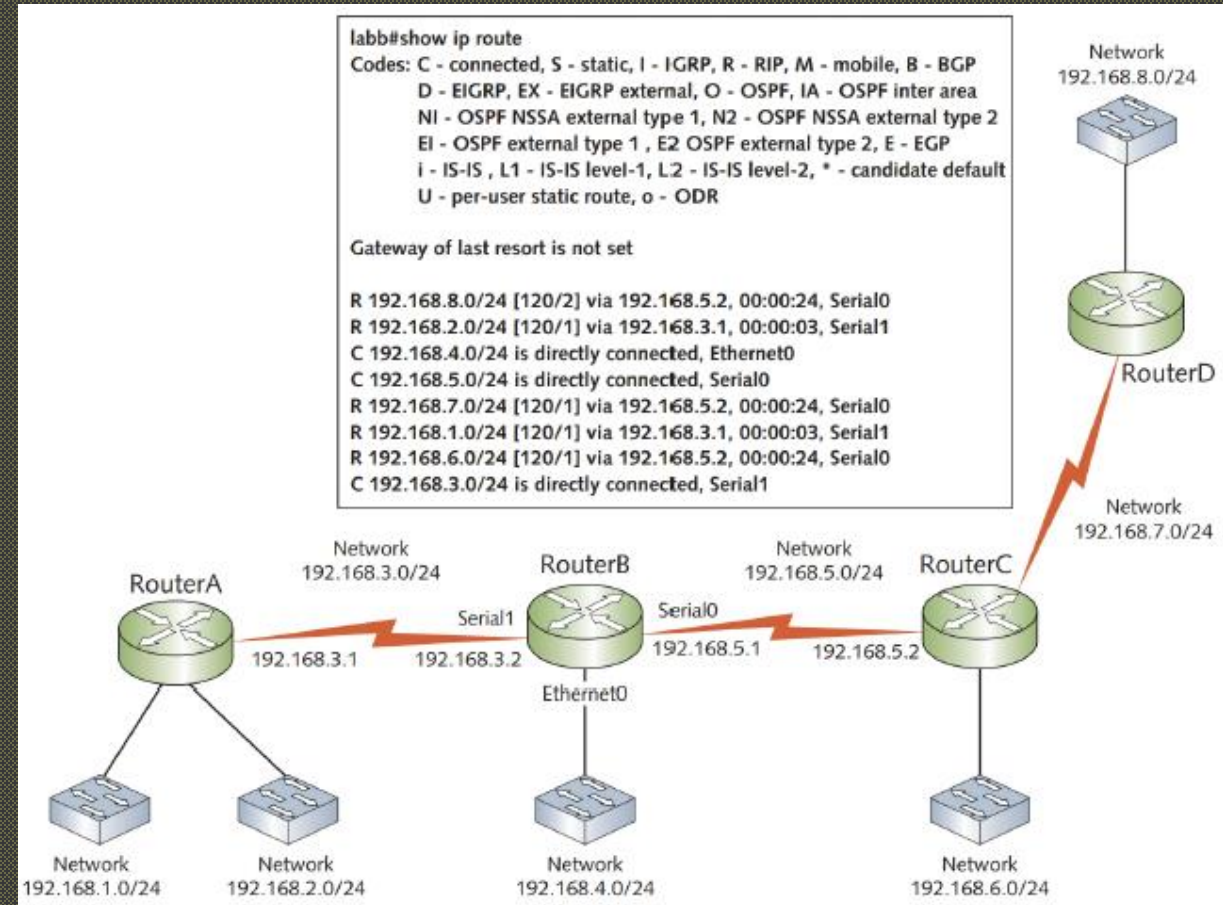
H1 Receives Reply: Closes the loop.



6.6 Routers

Routing Tables

- R - The route was learned through the RIP routing protocol.
- Destination Network: 192.168.8.0/24 - The target network for this route.
- Administrative Distance & Metric [120/2]:
 - 120 is the Administrative Distance (AD), which indicates the trustworthiness of the routing source. Routes with the lowest AD are preferred.
 - 2 is the metric (hop count), meaning the destination network is two hops away (via RouterC and RouterD).
- Next Hop: 192.168.5.2 - The IP address of the next router (RouterC) to which traffic will be forwarded.
- Age: 00:00:24 - The time since this route was last updated. A value consistently higher than 30 seconds may indicate a network issue.
- Outgoing Interface: Serial0 - The local router interface used to send packets to the next hop.



The table shown is the actual routing table of Router B

6.6 Routers

Routing Tables

- A routing table is like a map inside a router that helps it decide where to send data packets next. It tells the router *which path* to use to reach a particular destination network.

1. Destination Network

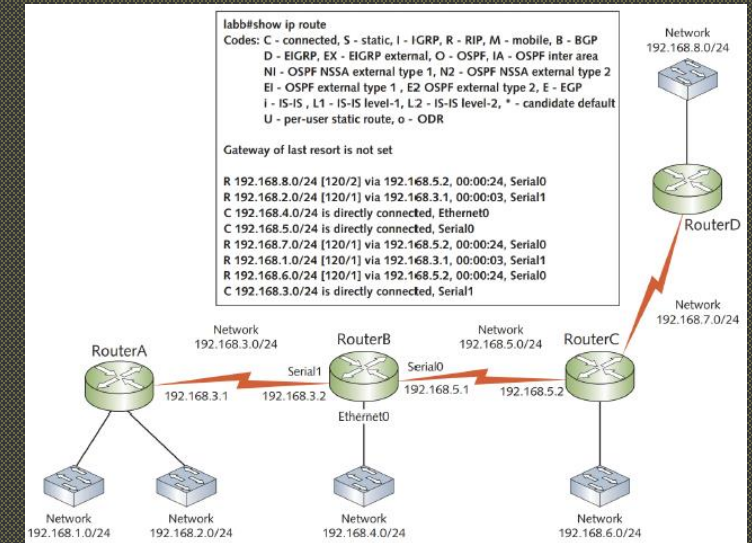
- This is the network address of the destination the router can send packets to.
- Written in CIDR notation (e.g., 172.16.0.0/16).
- When the router receives a packet, it checks the destination IP address inside that packet and compares it to its routing table.
- If a match is found → router knows which path (interface or next hop) to use.

2. Next Hop

- The next hop tells the router where to send the packet next.
- It can be:
 - A directly connected interface (e.g., Ethernet0 or FastEthernet1), or
 - The IP address of the next router (called a gateway).

3. Metric (Cost or Distance)

- The metric shows how “far” or “expensive” a route is.
- It is a numeric value used to choose the best path when multiple routes exist.
- Lower metric = better path.



6.6 Routers

Routing Tables

4. Routes get into the table in three main ways:

- Directly Connected Route:

- The network is connected directly to the router interface.
- Example: Router interface GigabitEthernet0/1 has IP 192.168.1.1/24 → this network (192.168.1.0/24) is directly connected.

- Static Route:

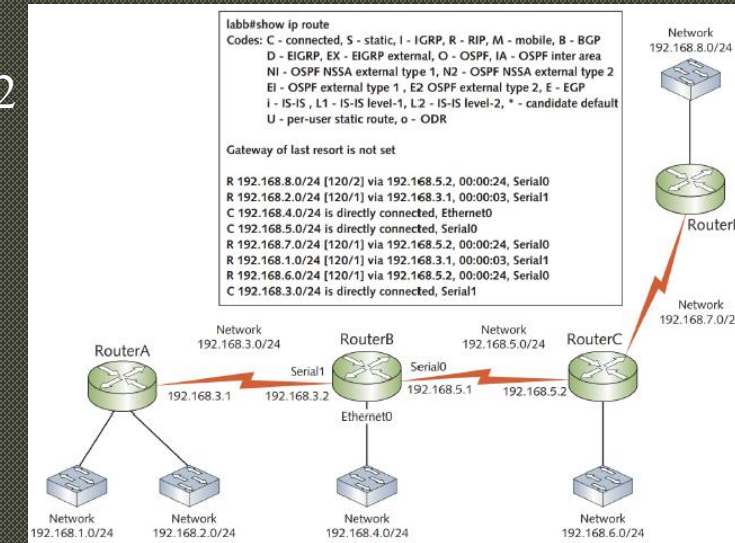
- Manually added by an administrator.
- Example: `ip route 10.0.0.0 255.255.255.0 192.168.5.2`
- This tells the router to send packets for 10.0.0.0/24 to the next router at 192.168.5.2

- Dynamic Route:

- Learned automatically using a routing protocol (like RIP, OSPF, or EIGRP).
- The router communicates with others and updates routes automatically.

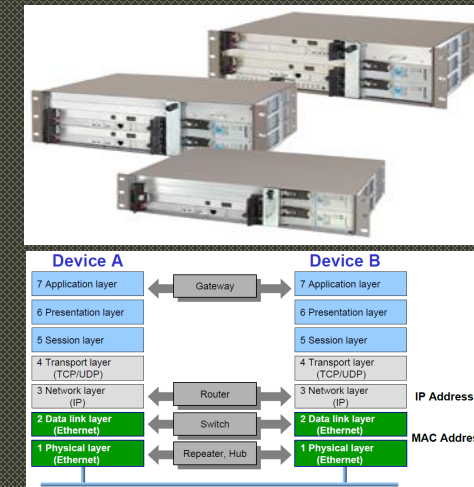
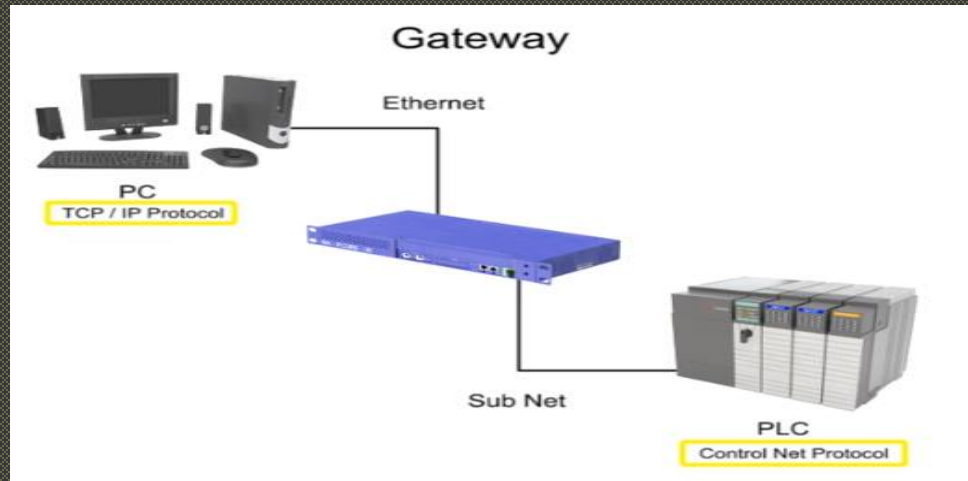
5. Timestamp

- Used only for dynamic routes.
- Shows how recently a route was updated.
- Helps detect stale or outdated routes.



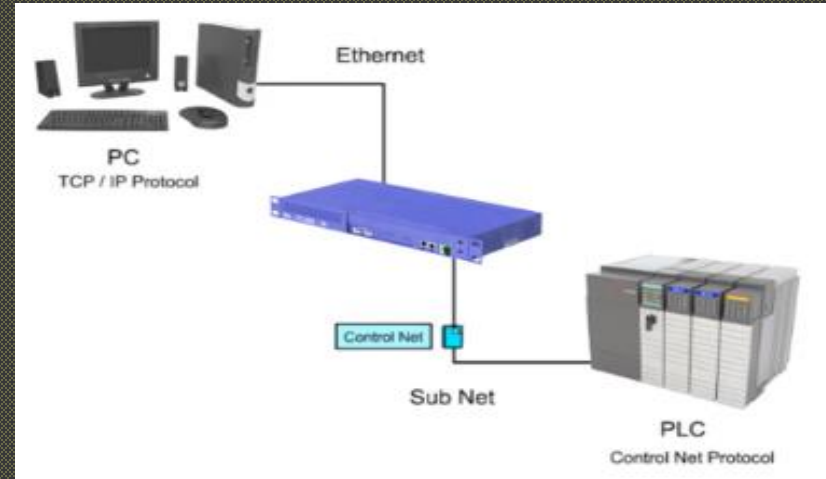
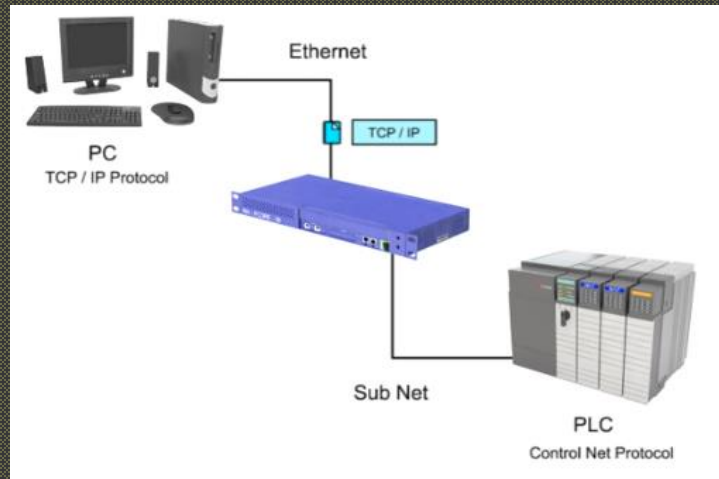
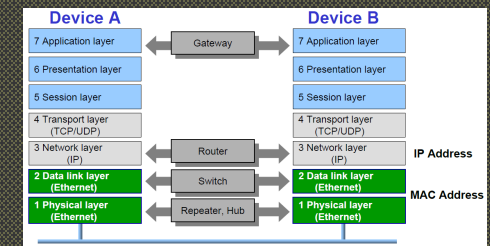
6.7 Gateways

- A gateway is a network node (either a hardware device, software application, or a combination of both) that serves as an entry/exit point between two or more networks that use different communication protocols, architectures, or data formats.
- When packets arrive at a gateway, the software strips all networking information from the packet, leaving only the raw data.
- The gateway then translates the data into the new format and sends it back down the OSI layers using the networking protocols of the destination system.
- Because gateways translate data, they generally operate at the upper layers of the OSI model. Usually, this takes place at the Application layer.



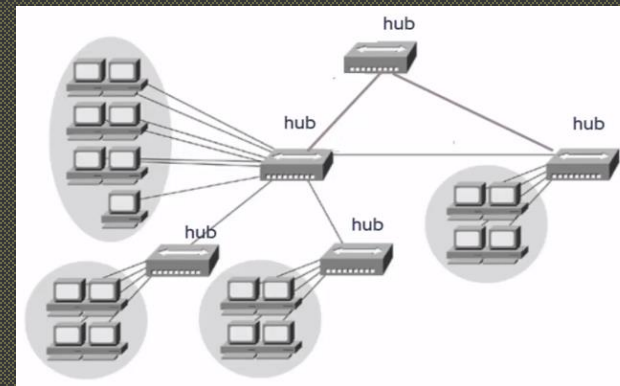
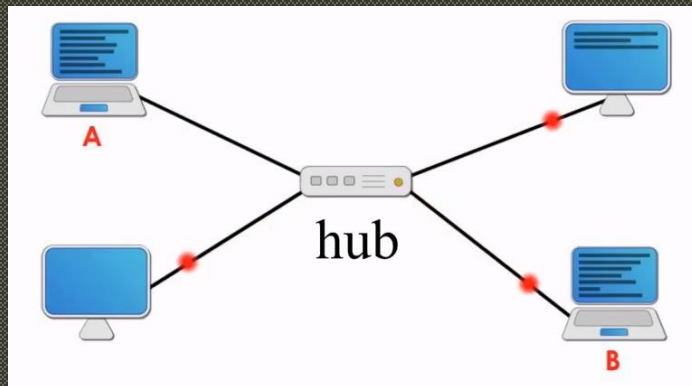
6.7 Gateways

- Gateways are also called protocol converters and can operate at any network layer.
- Gateways perform a conversion function so that a computer in Ethernet network using a TCP/IP protocol suite may communicate with a PLC on a Sub Net using another incompatible protocol (for example, ControlNet protocol in the figure).



6.8 Collision and Broadcast Domain

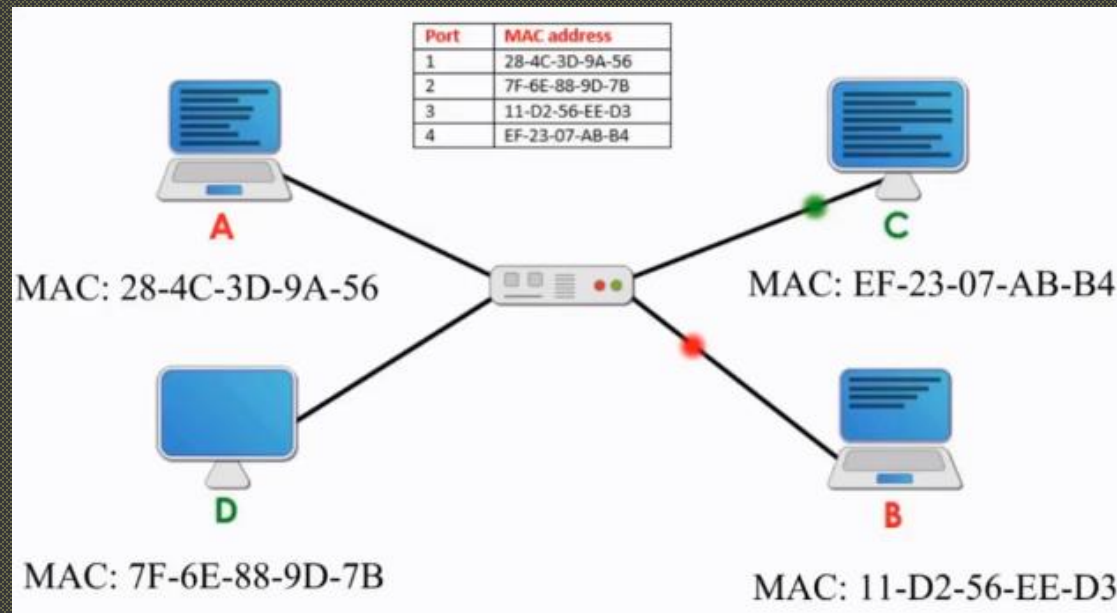
- In an Ethernet network, a collision domain is a group of nodes that can hear each other. Collision occurs when two nodes send signals at the same time.
- Broadcast domain is a group of nodes that can reach/hear each other by broadcasting (using MAC address FF-FF-FF-FF-FF-FF as destination address)
- When a **hub** is used:
 - it creates and expands a collision and a broadcast domain
 - all the nodes connected to it create one collision domain that share the bandwidth of the network
 - more hubs are used in a LAN, the larger is the collision domain



6.8 Collision and Broadcast Domain

When a **switch** is used:

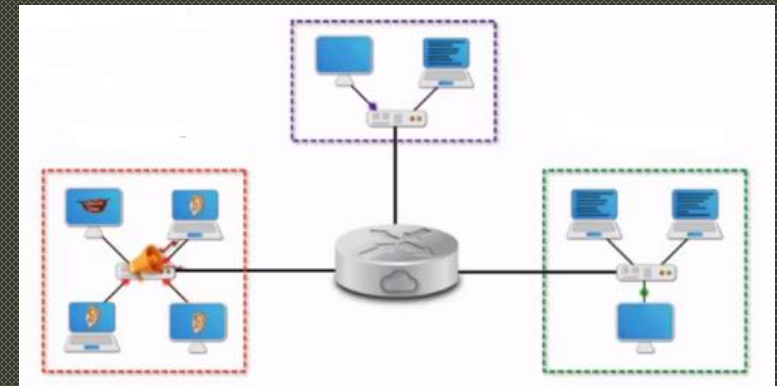
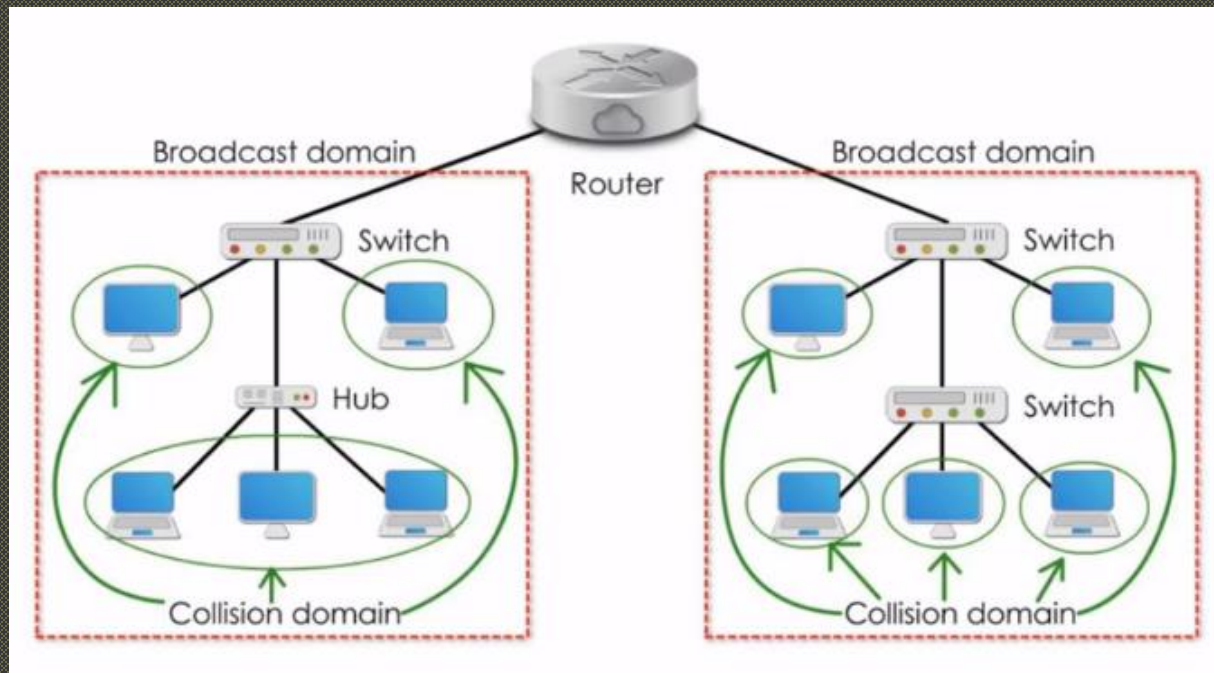
- it creates one-to-one conversations with no collisions
- it gives each conversation the full bandwidth of the network
- it separates collision domains
- it creates broadcast domain



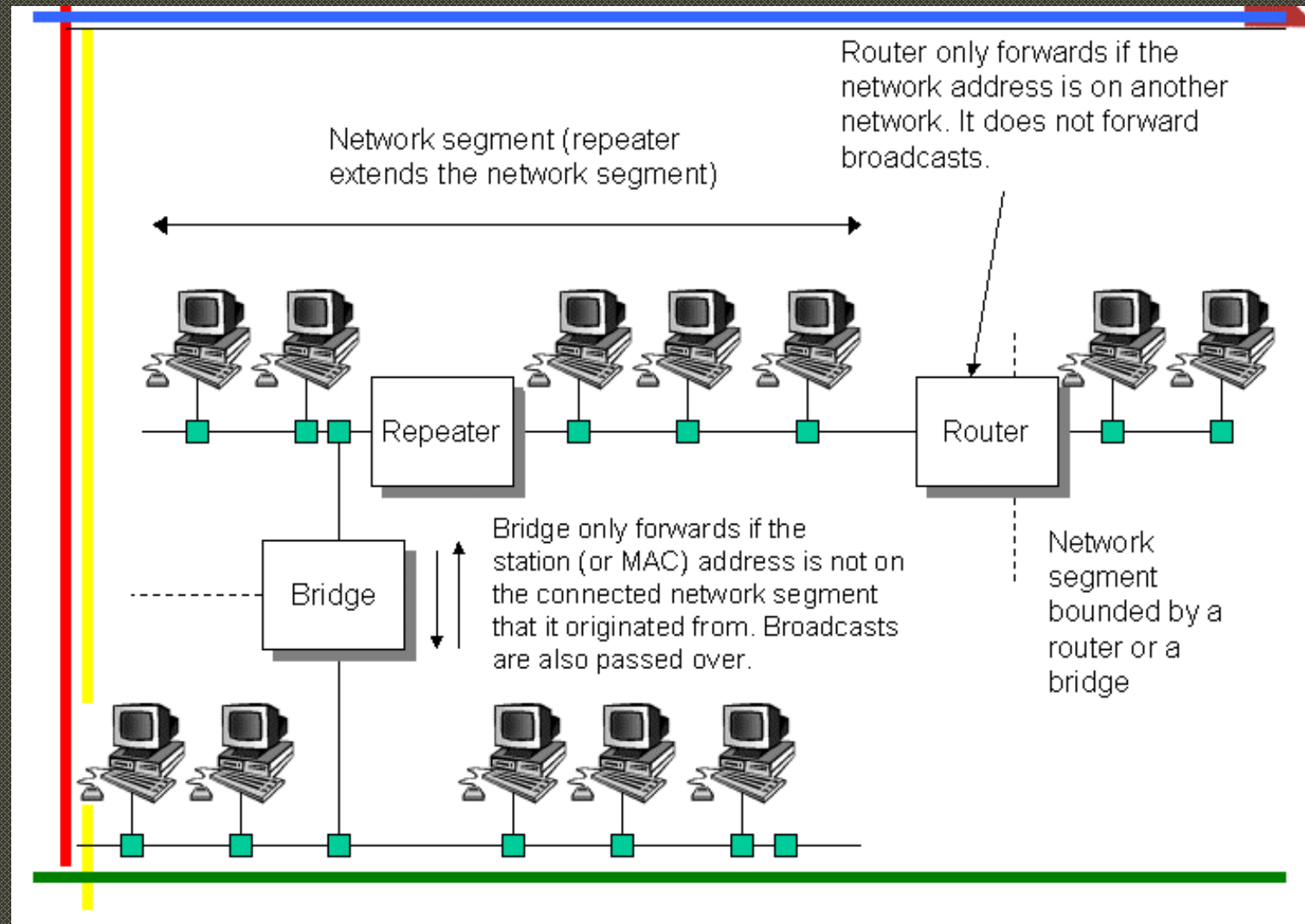
6.8 Collision and Broadcast Domain

When a **router** is used:

- it forwards IP packets based on IP address across LANs
- it separates collision domains
- it separates broadcast domain; it will not forward broadcasting packets across the LANs

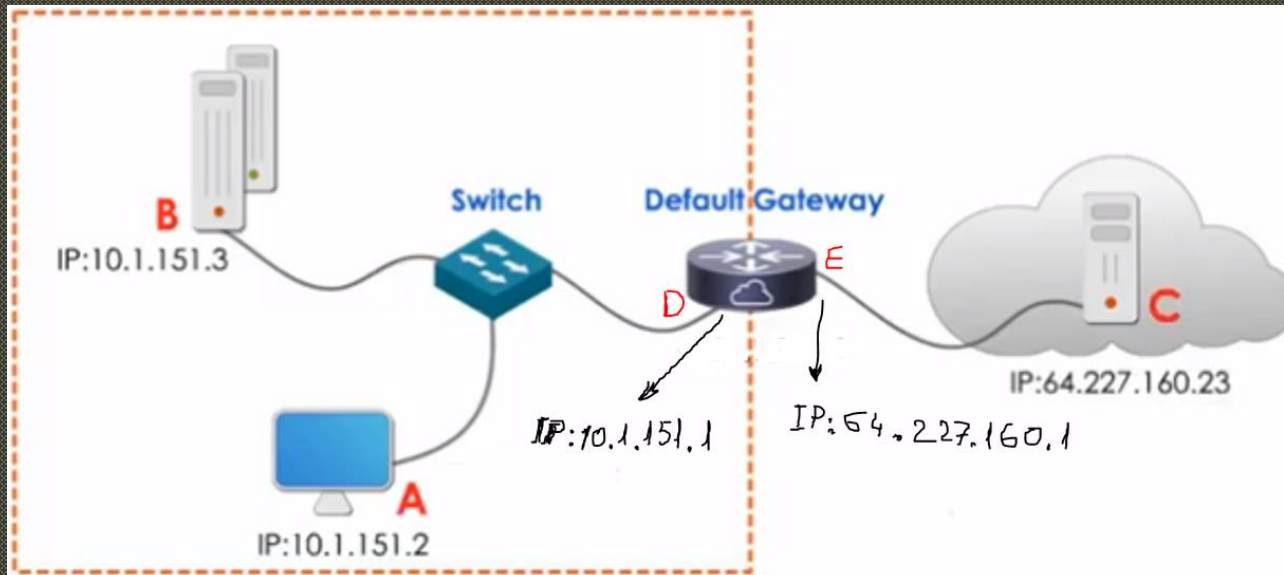


6.8 Repeaters, Bridges and Routers in LANs



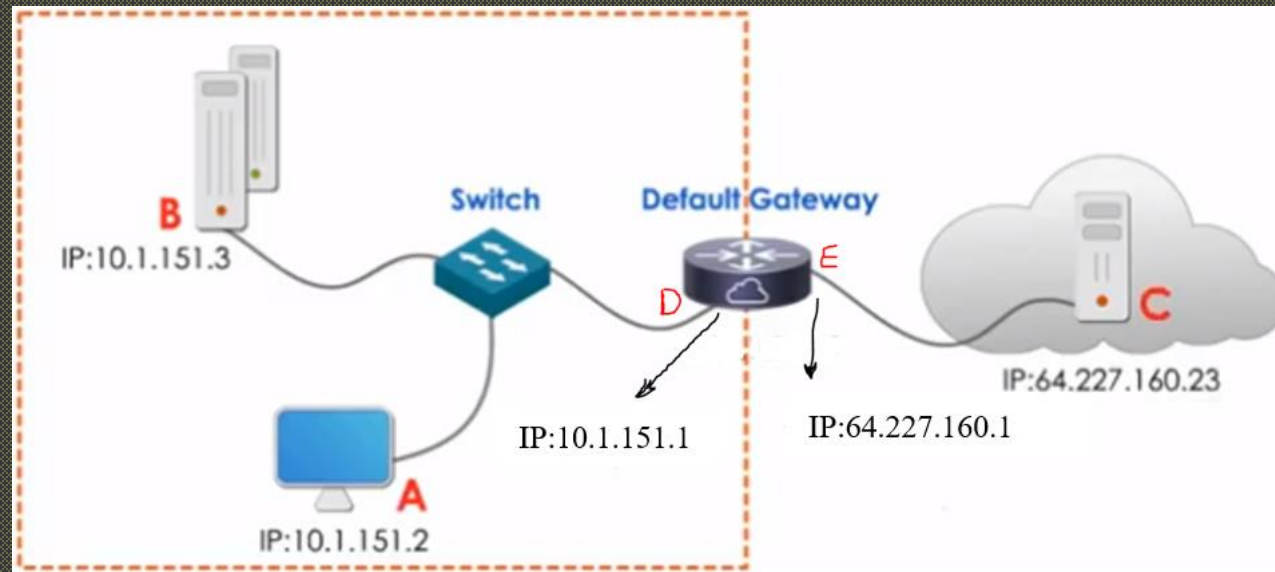
6.9 Local vs Remote Communications

- When station A communicates with station B, it knows that station B is in the same LAN by looking at the IP address of station B and subnet mask.
- Once station A knows that station B is in the same LAN, it uses ARP (the program located in station A) to get the MAC address of station B based on IP address of station B (letters A, B, C, D, and E represent MAC addresses).
- Station A uses the MAC address of station B to locally communicate with station B through the layer 2 switch.



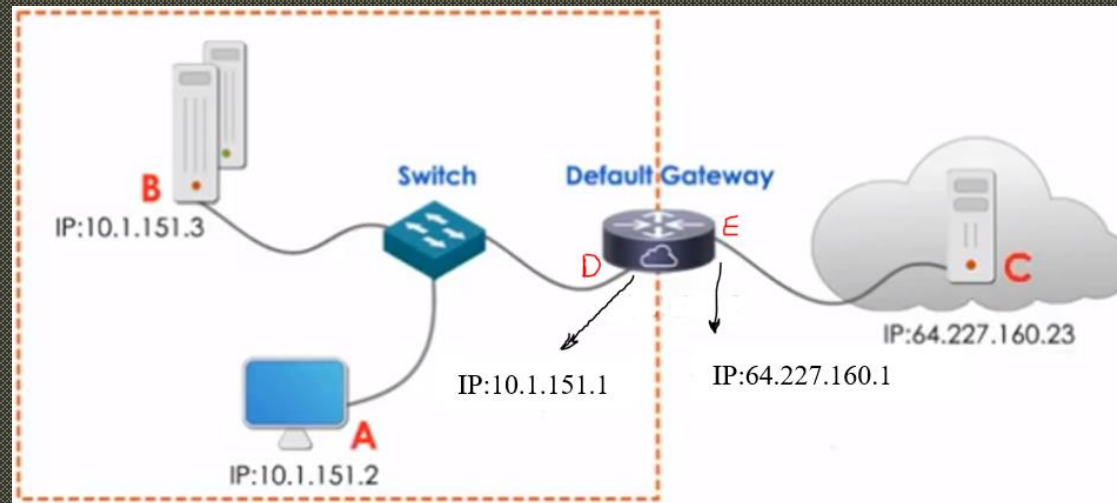
6.9 Local vs Remote Communications

- When station A communicates with station C from one LAN to another LAN, it is called remote communication.
- When station A finds out that station C is in another LAN (based on IP address of station C and subnet mask), it sends an ARP request to find the MAC address of the router (Default Gateway) based on IP address of the router.
- Once the MAC address (D) of the router is known, station A sends the packet to the router.
- In the same way, the router will deliver the packet to station C by finding the MAC address of station C through another ARP request on the other port of the router (E).



6.9 Local vs Remote Communications

- Subnet mask is used to determine the network of the target station (if it is in the same LAN or different LAN from the source station).
- MAC address is used for local communications (in the same LAN).
- IP address is used for remote communications (between different LANs).
- A layer 2 switch is used for local communications.
- A layer 3 router is used for remote communications.
- The switch does not have MAC addresses on any of its ports. It uses the MAC address of the stations connected in each of its ports.
- The router has MAC addresses and IP addresses on each of its ports. IP addresses on each port match with the LAN addresses.



6.10 Firewall

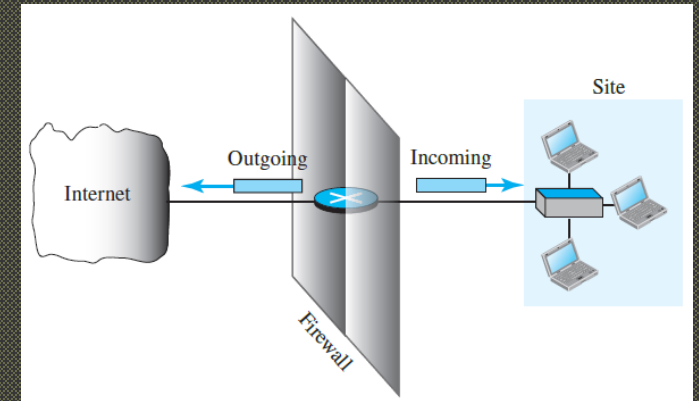
- A firewall is a security device (can be a router, computer, or specialized hardware) placed between an internal network (like a company's LAN) and the Internet.
- It acts like a security guard - it checks every piece of data (called a packet) that tries to go in or out of the network and decides whether to allow it or block it.

Why is needed:

- Other security tools (like encryption or authentication) cannot stop all attacks.
A firewall helps by controlling which traffic is allowed to reach your network or leave it.
- Example:
 - It can block all incoming emails from a suspicious IP address.
 - Or stop employees from visiting social media websites during work hours.

How It Works

- A firewall checks packets according to a set of rules (called a filtering table or policy). Some packets are forwarded (allowed), others are filtered (blocked).



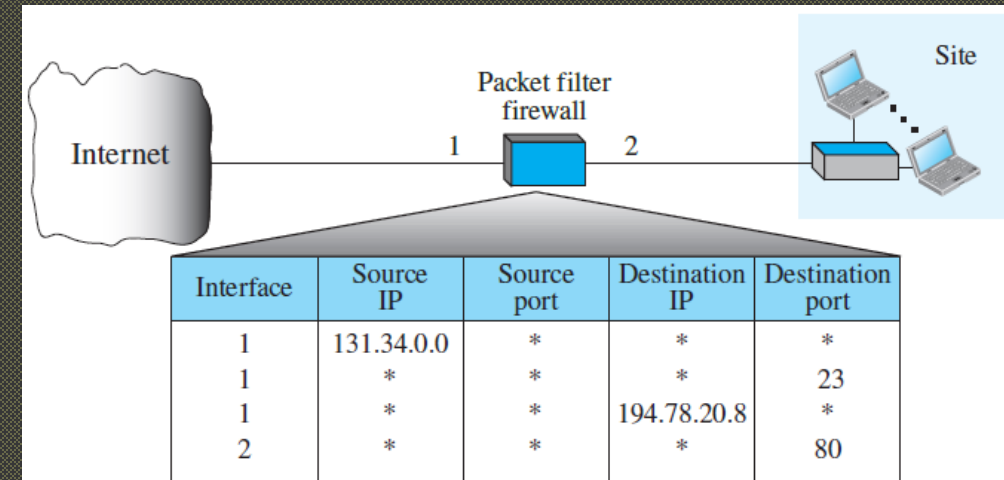
6.10 Firewall

Two Main Types of Firewalls

- Packet-Filter Firewall
- Proxy Firewall (Application Gateway)

1. Packet-Filter Firewall

- Works at the Network Layer (IP) and Transport Layer (TCP/UDP).
- Makes decisions based on header information in packets:
 - Source IP address – who sent the data
 - Destination IP address – where the data is going
 - Source port number – the port used by the sender
 - Destination port number – the port used by the receiver
 - Protocol – whether it's TCP, UDP, etc.
- How It Works
 - It uses a filtering table - like a checklist - to decide what to do with each packet.



Rule	Description	What Happens
1	Block all packets from network 131.34.0.0	Stops suspicious traffic from that entire network
2	Block all packets going to port 23 (TELNET)	Prevents remote logins into company computers
3	Block all packets destined for host 194.78.20.8	Keeps this internal host private
4	Block all packets going out to port 80 (HTTP)	Prevents employees from browsing websites

6.10 Firewall

2. Proxy Firewall (Application Gateway)

- Works at the Application Layer (the top layer of communication).
- Examines the actual content of messages - not just headers.
- Can make smarter decisions based on what the user is trying to do.

Why It is Needed

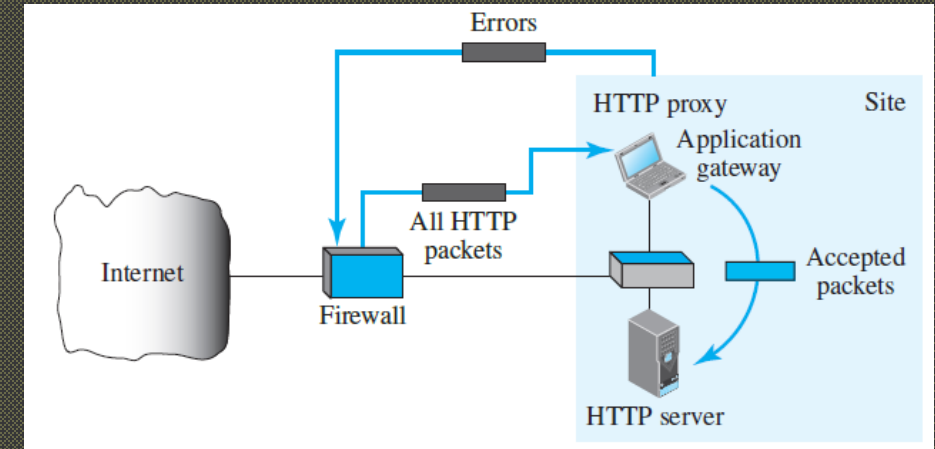
- Sometimes, filtering only by IP or port is not enough.

For example:

- All web traffic uses port 80 (HTTP), but not all users should access all web pages.
- A company might want to allow access only to clients who have business relationships.
- A packet filter cannot make this distinction - it only sees “port 80 traffic.”
A proxy firewall can - because it reads the application data (like URLs, HTTP headers, etc.).

How It Works

- The user's computer sends a request to the proxy firewall (not directly to the web server).
- The proxy (application gateway) checks the request:
 - Is this user allowed?
 - Is this website or URL permitted?
- If approved, the proxy opens a connection to the real web server and forwards the request.
- If not approved, it sends an error message back to the user.



References

- Guide to Networking Essentials, 8th Edition, Greg Tomsho
- Data Communications and Networking with TCP/IP Protocol Suite, 6th Edition, Behrouz A. Forouzan
- Sunny Classroom (youtube.com)
- <https://www.networkacademy.io/courses>

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Thank you!

Discussions?